LAB 2

AGILE BACKLOG CREATION & SPRINT SIMULATION IN JIRA

Municipal Solid Waste Route Optimizer

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1. Lab Objective

Create and prioritize an Agile backlog in Jira, estimate stories using story points, organize work into one-week sprints, simulate Scrum progress, and monitor progress using Jira's Burndown Chart.

2. Project Scenario

The Municipal Solid Waste Route Optimizer receives ultrasonic smart-bin fill information, generates optimized garbage-collection routes, supports truck drivers, and tracks missed-pickup citizen complaints. Primary actors are the Sanitation Supervisor and Truck Driver.

3. Epics Created

Key	Epic Name
SCRUM-11	Smart Bin Monitoring
SCRUM-12	Route Optimization
SCRUM-13	Truck & Driver Management
SCRUM-14	Missed Pickup Complaint Management
SCRUM-15	Sanitation Administration
SCRUM-16	Performance & Security

4. User Stories, Priority & Story Points

Key	User Story	Epic	Pts	Priority
SCRUM-17	Monitor Smart Bin Fill Levels	Smart Bin Monitoring	5	High
SCRUM-18	Identify Overfilled Bins	Smart Bin Monitoring	3	High
SCRUM-19	View Bin Status	Smart Bin Monitoring	3	Medium
SCRUM-20	Generate Optimized Collection Routes	Route Optimization	8	High
SCRUM-21	Include High-Fill Bins in Routes	Route Optimization	5	High
SCRUM-22	Recalculate Collection Routes	Route Optimization	8	High
SCRUM-23	Assign Routes to Truck Drivers	Truck & Driver Management	5	High
SCRUM-24	View Assigned Collection Route	Truck & Driver Management	3	High
SCRUM-25	Update Collection Status	Truck & Driver Management	3	Medium
SCRUM-26	Register Missed Pickup Complaint	Missed Pickup Complaint Management	3	High
SCRUM-27	Track Complaint Status	Missed Pickup Complaint Management	3	Medium
SCRUM-28	Resolve Missed Pickup Complaint	Missed Pickup Complaint Management	5	High
SCRUM-29	Manage Trucks	Sanitation Administration	5	Medium
SCRUM-30	Manage Smart Bins	Sanitation Administration	5	Medium
SCRUM-31	Monitor Collection Operations	Sanitation Administration	5	Medium
SCRUM-32	Fast Route Recalculation	Performance & Security	8	High
SCRUM-33	Secure System Access	Performance & Security	5	High
SCRUM-34	Handle Large-Scale Bin Data	Performance & Security	8	High

Total: 18 user stories and 90 story points.

5. Planning Poker / Estimation

Planning Poker was used as the estimation concept for discussing relative effort, complexity, risk and uncertainty. The selected story-point values were recorded in Jira.

6. Sprint Planning

Sprint	Dates	Stories	Points
Sprint 1	1–7 Sep 2026	SCRUM-17, SCRUM-18, SCRUM-19, SCRUM-20, SCRUM-21, SCRUM-23, SCRUM-24	32
Sprint 2	8–14 Sep 2026	SCRUM-22, SCRUM-25, SCRUM-26, SCRUM-27, SCRUM-28, SCRUM-29, SCRUM-30, SCRUM-31, SCRUM-32, SCRUM	58

7. Sprint Execution & Results

Sprint 1: 7/7 stories completed, 32/32 points. Sprint 2: 11/11 stories completed, 58/58 points. Overall: 18/18 stories and 90/90 points completed.

8. Burndown Charts

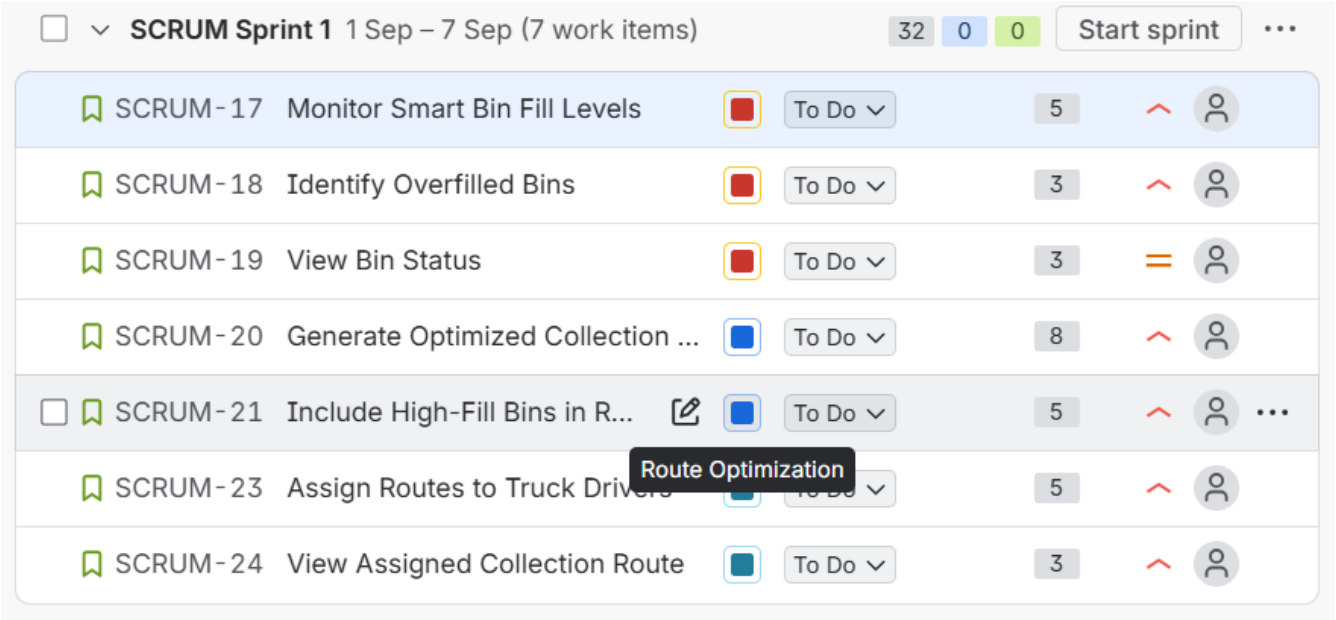
Jira's Sprint Burndown Chart was generated using Story points. The red line represents remaining work and the grey line represents the ideal burn guideline. Sprint 1 started at 32 points and reached 0; Sprint 2 started at 58 points and reached 0.

9. Comments

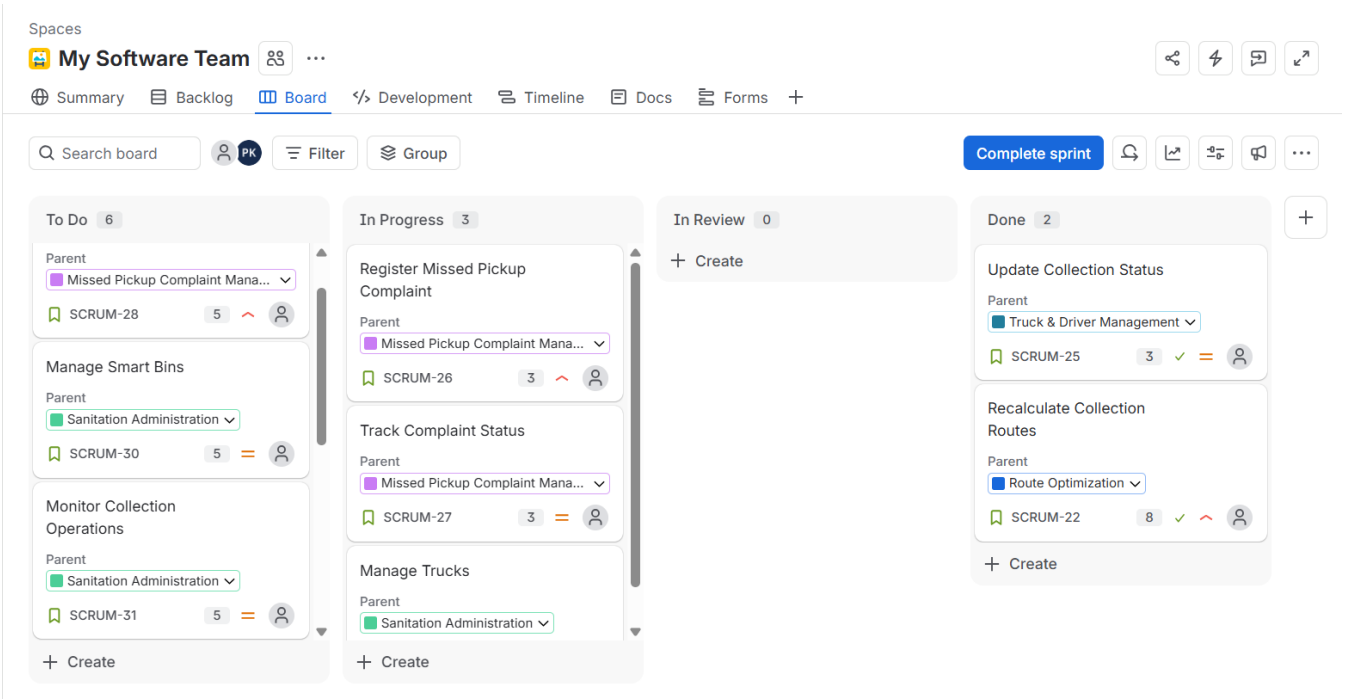
A Jira comment screenshot has not been fabricated. Before submission, add a real comment to one completed story and capture it. Suggested comment: “Implemented and verified the story during the mock sprint. Marked as Done after completion.”

10. Jira Evidence Screenshots

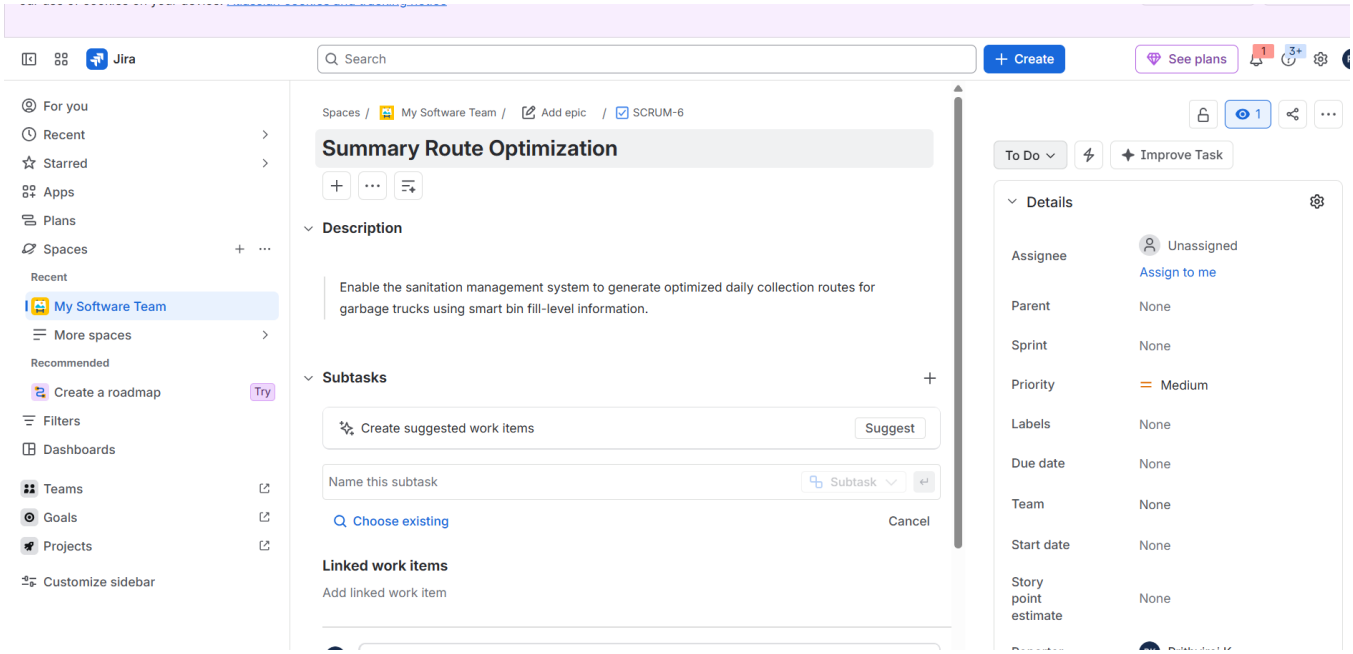
Available Jira screenshots included: 42.



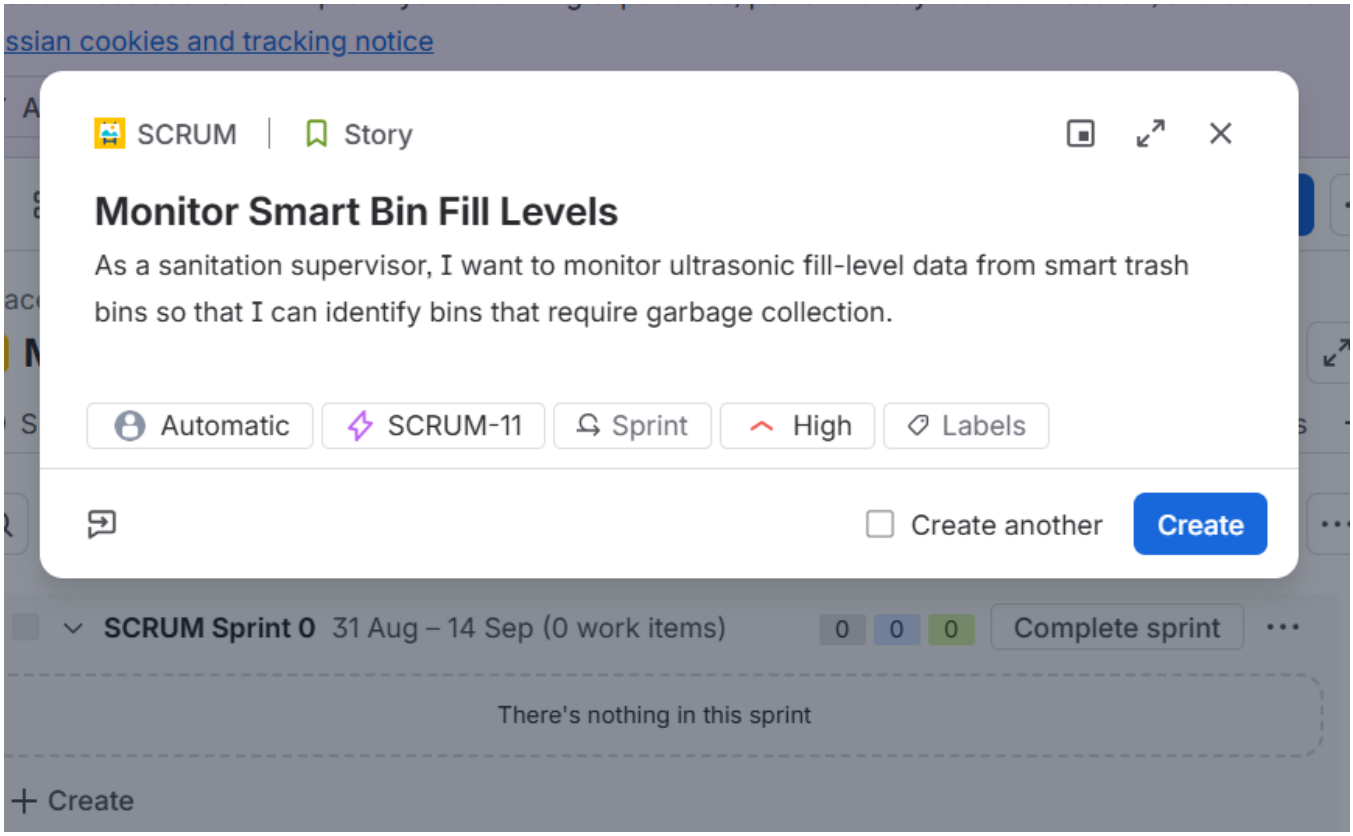
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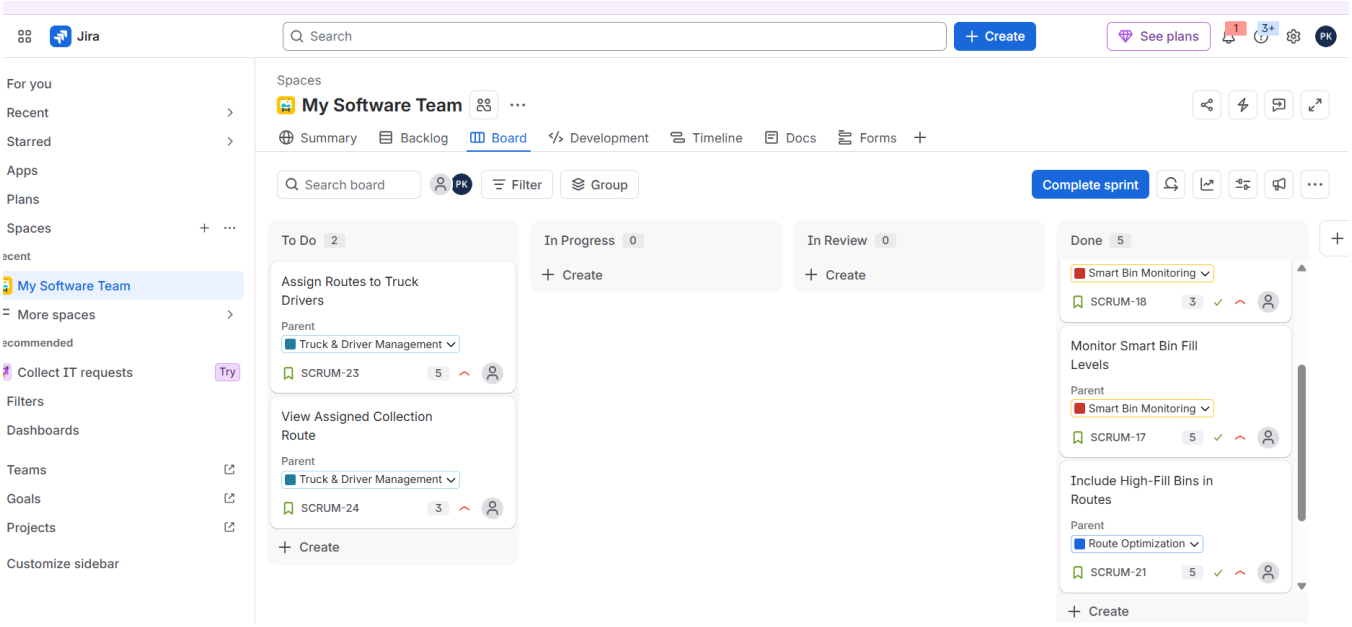
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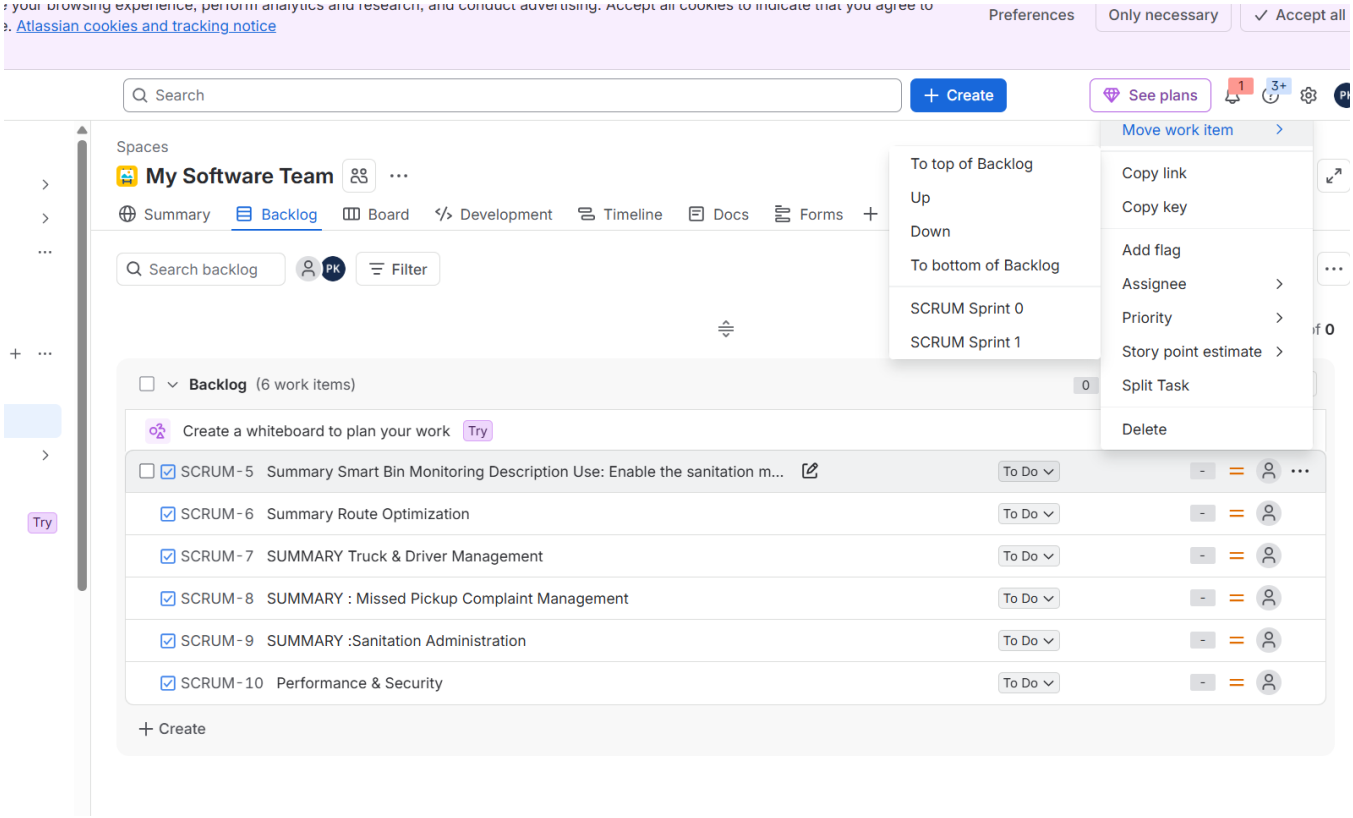
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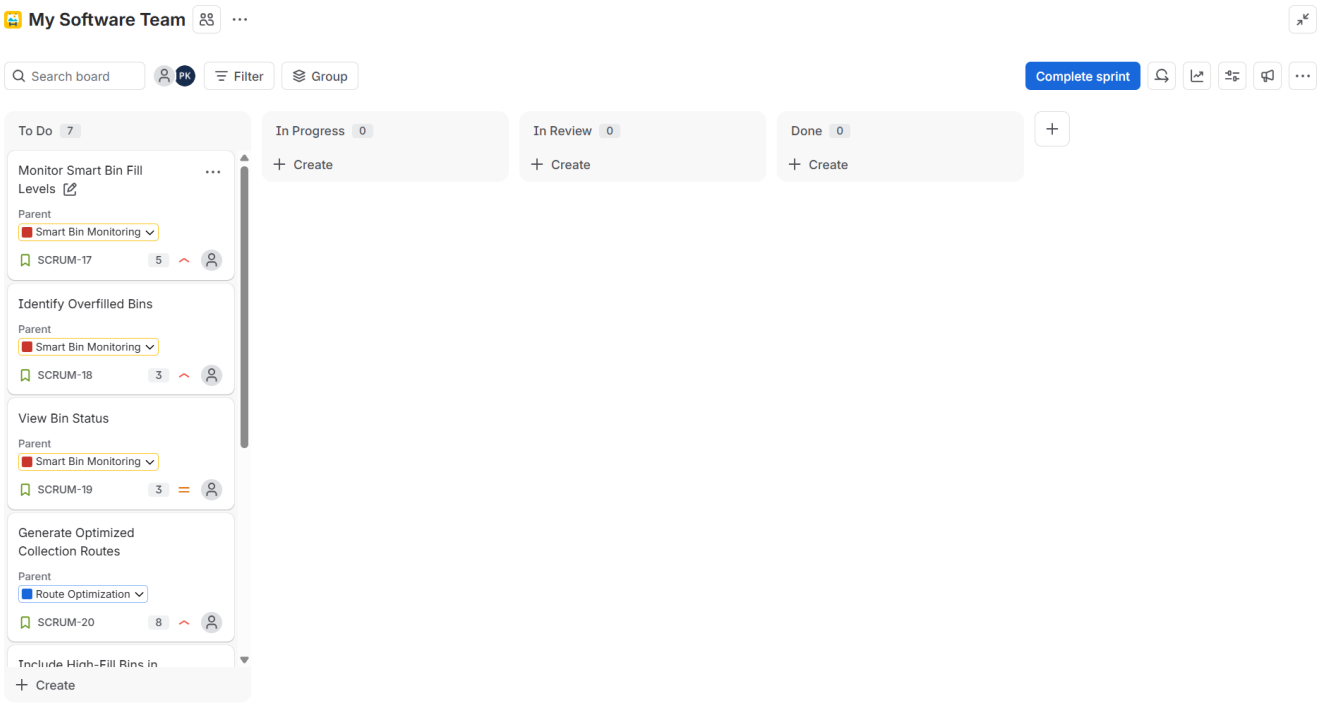
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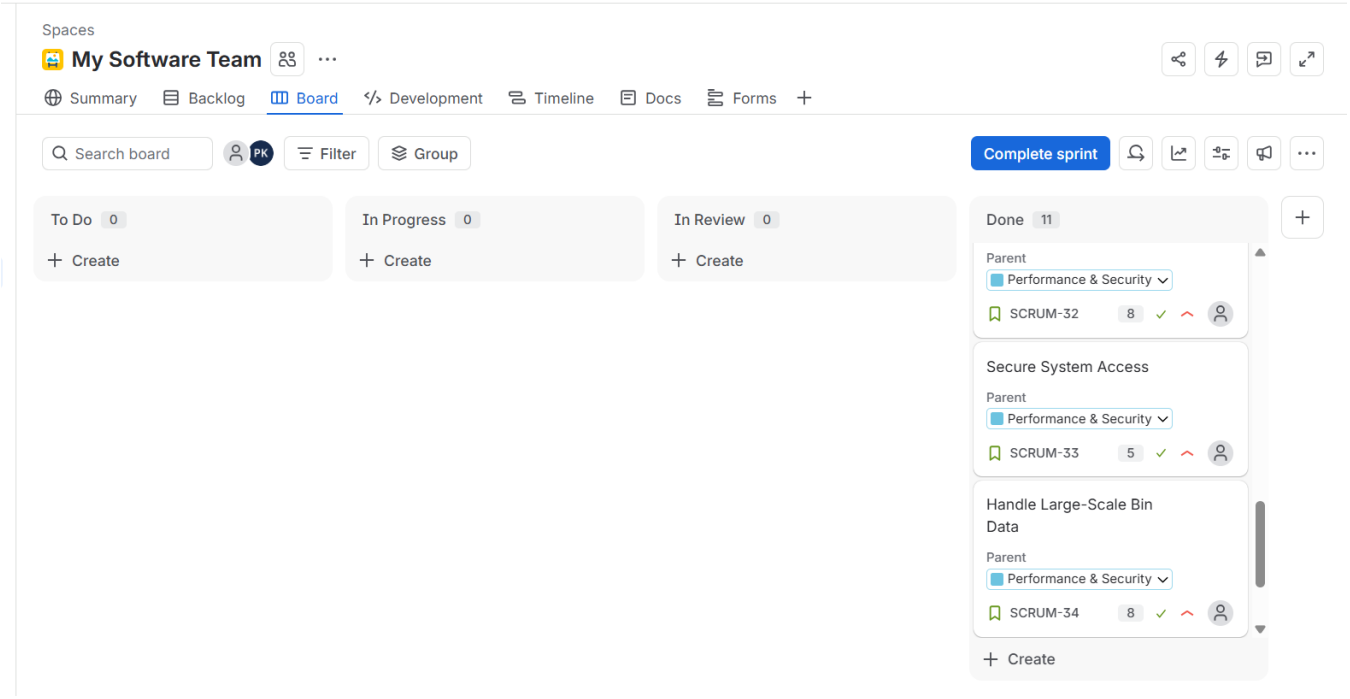
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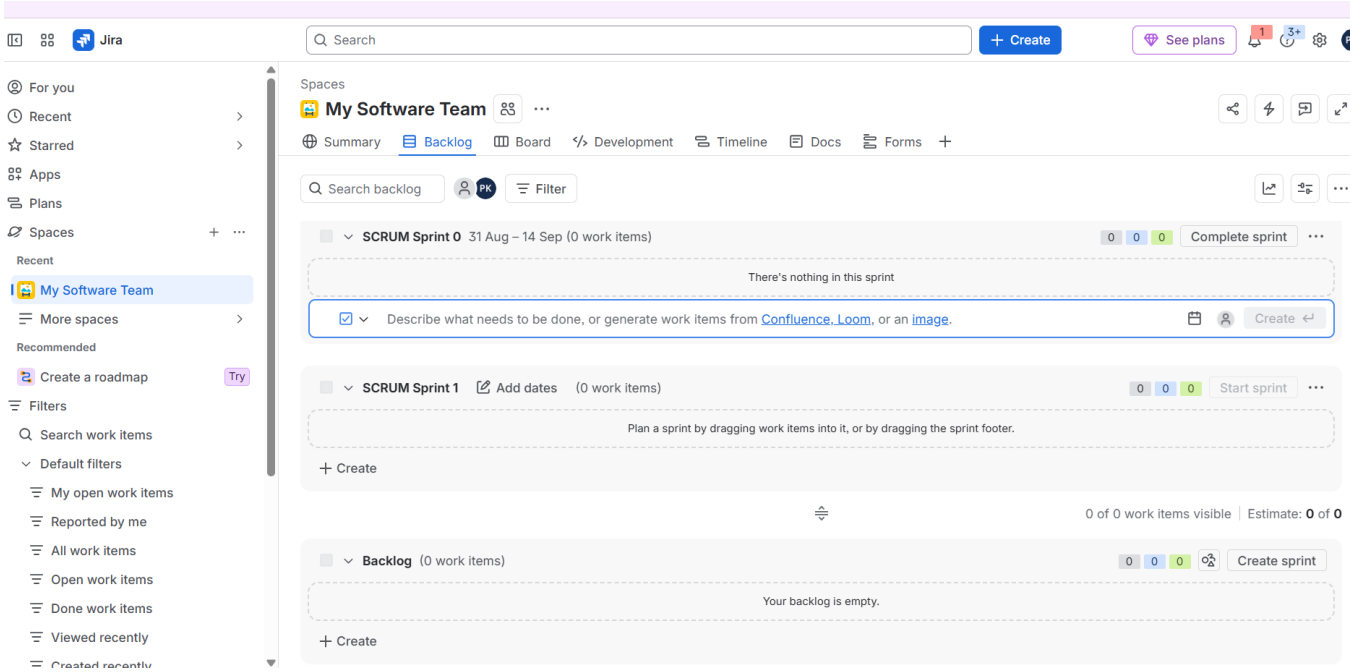
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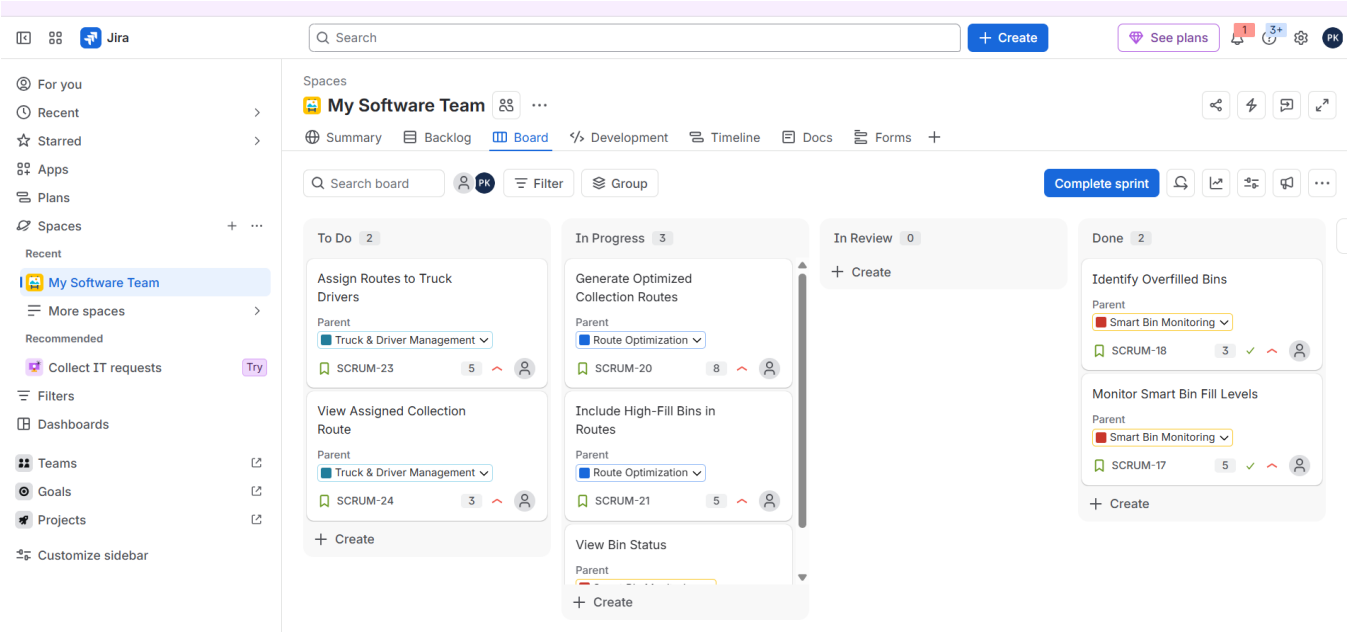
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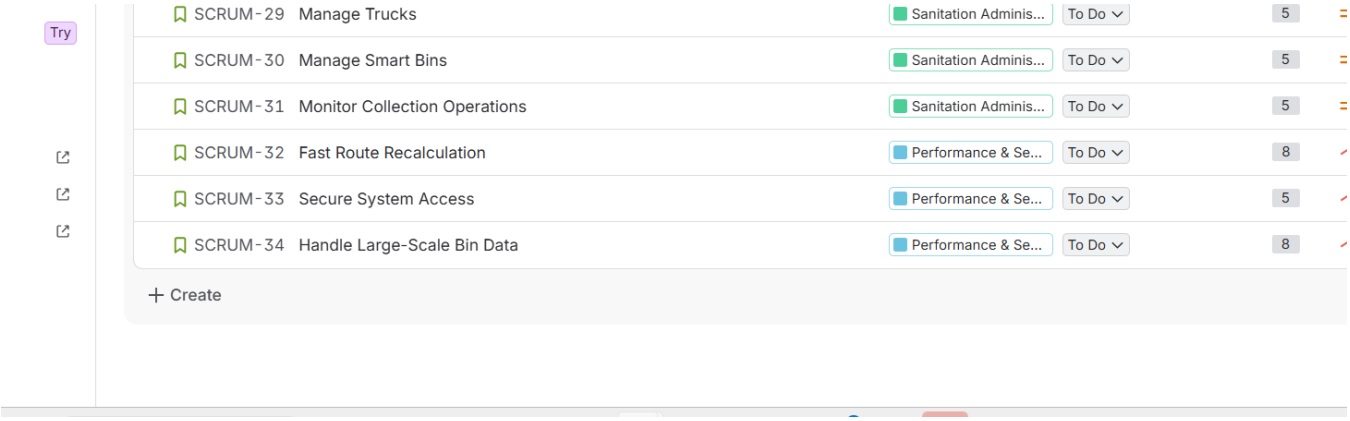
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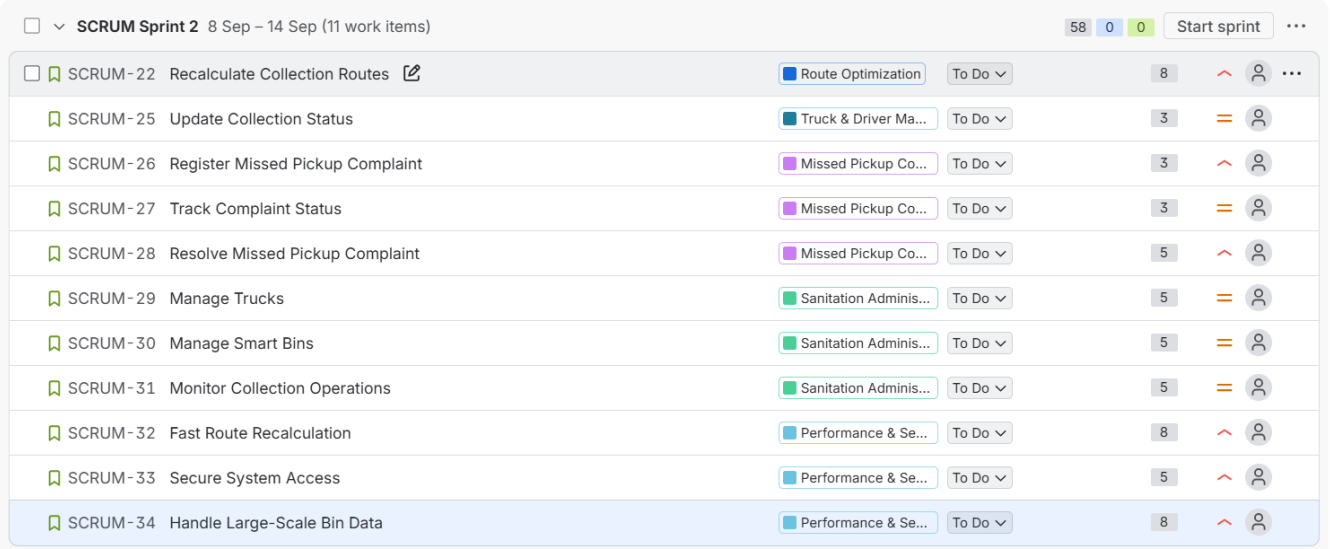
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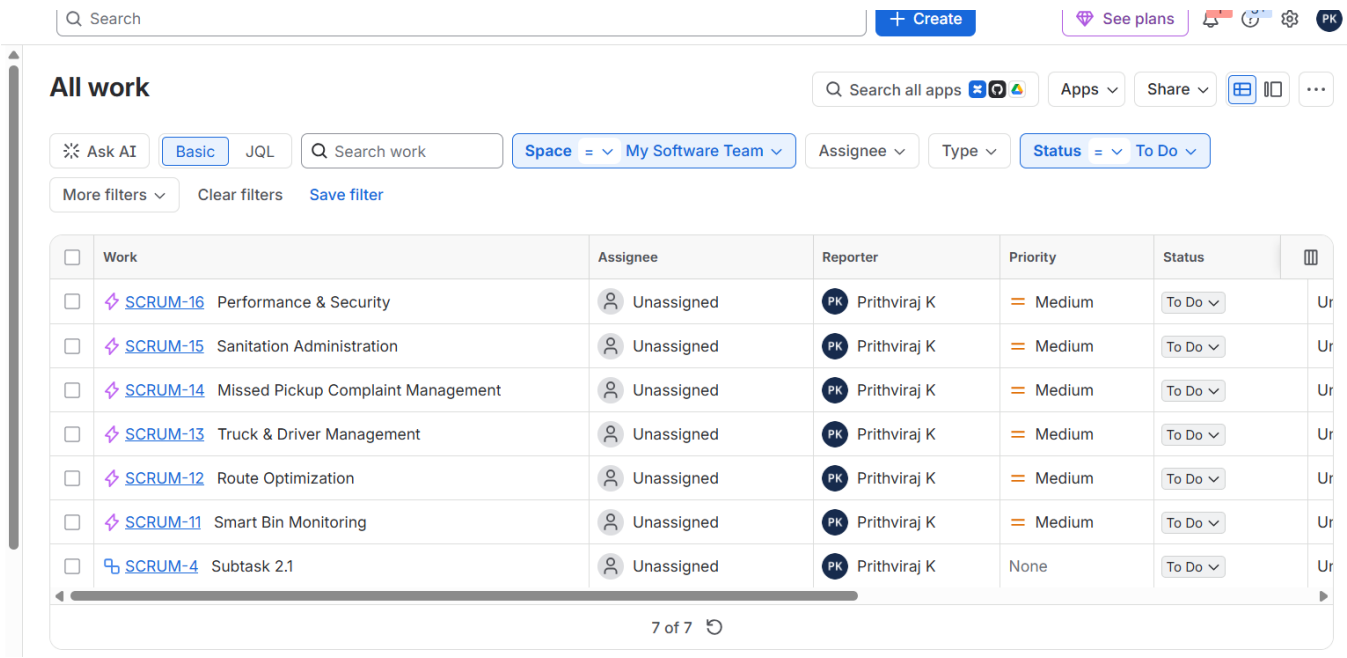
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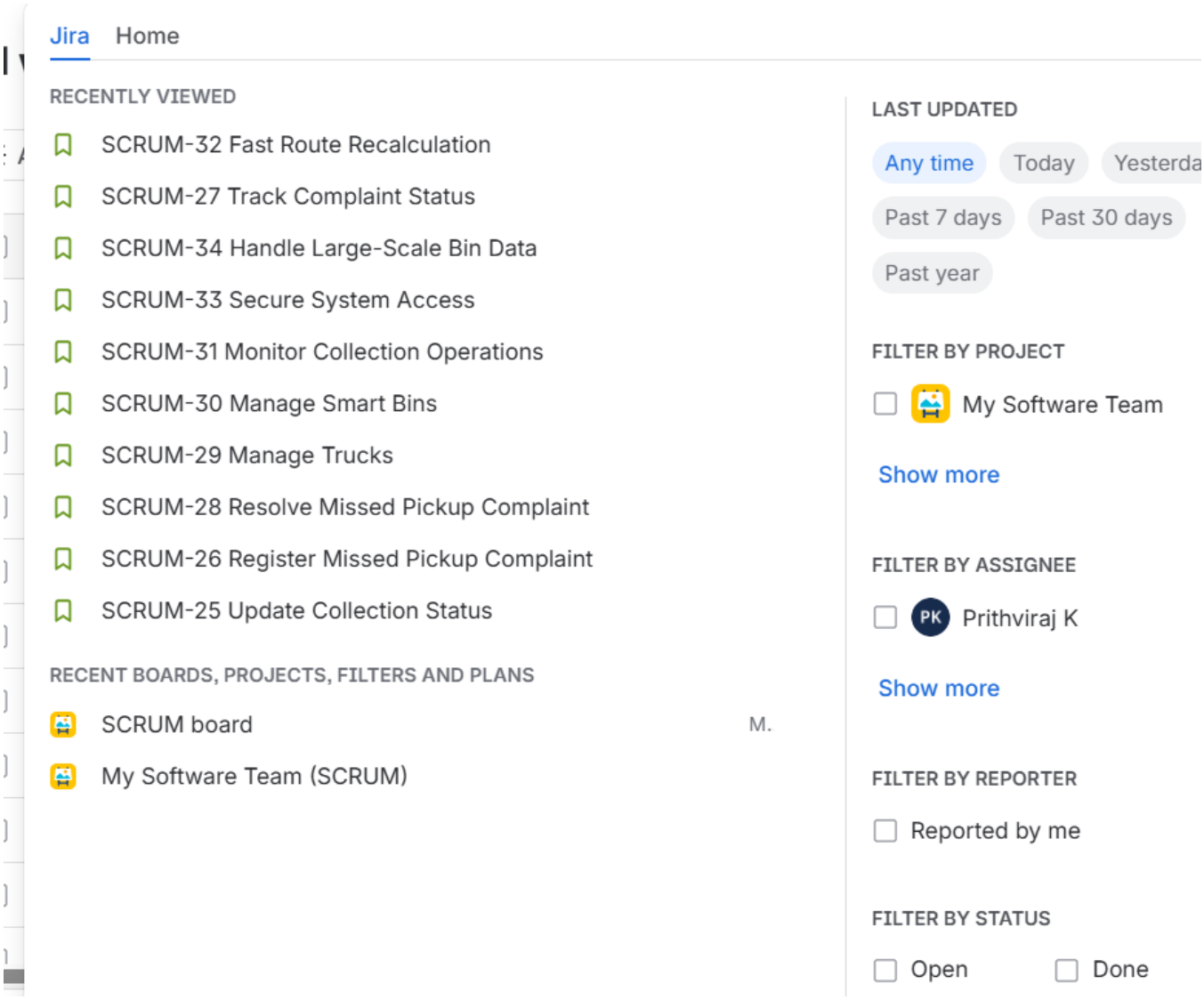
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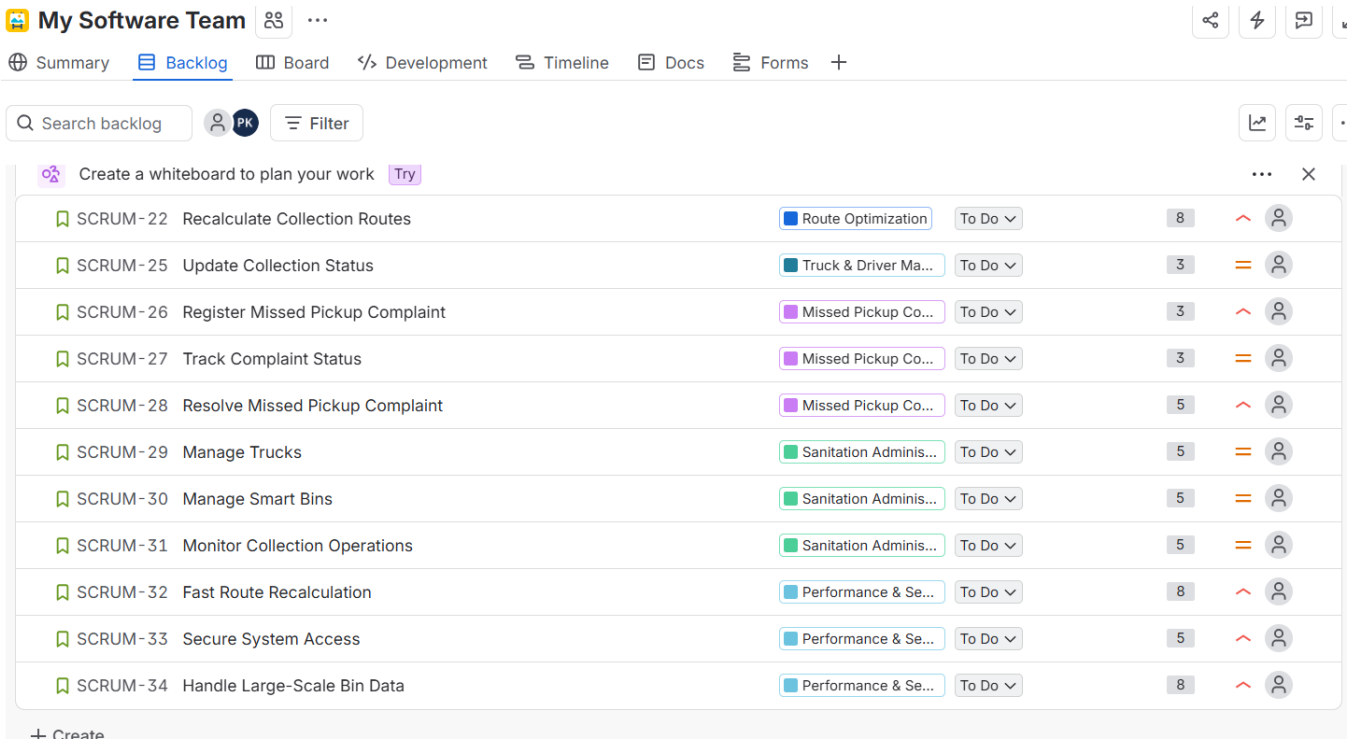
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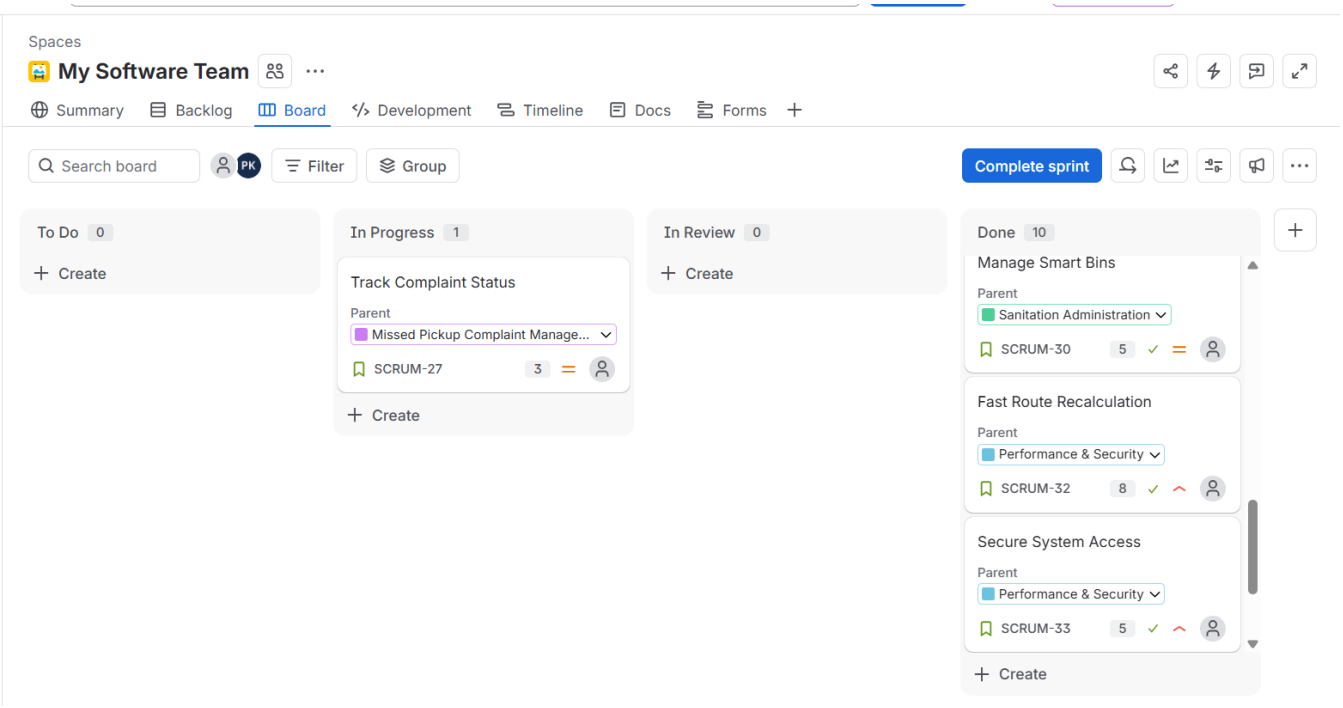
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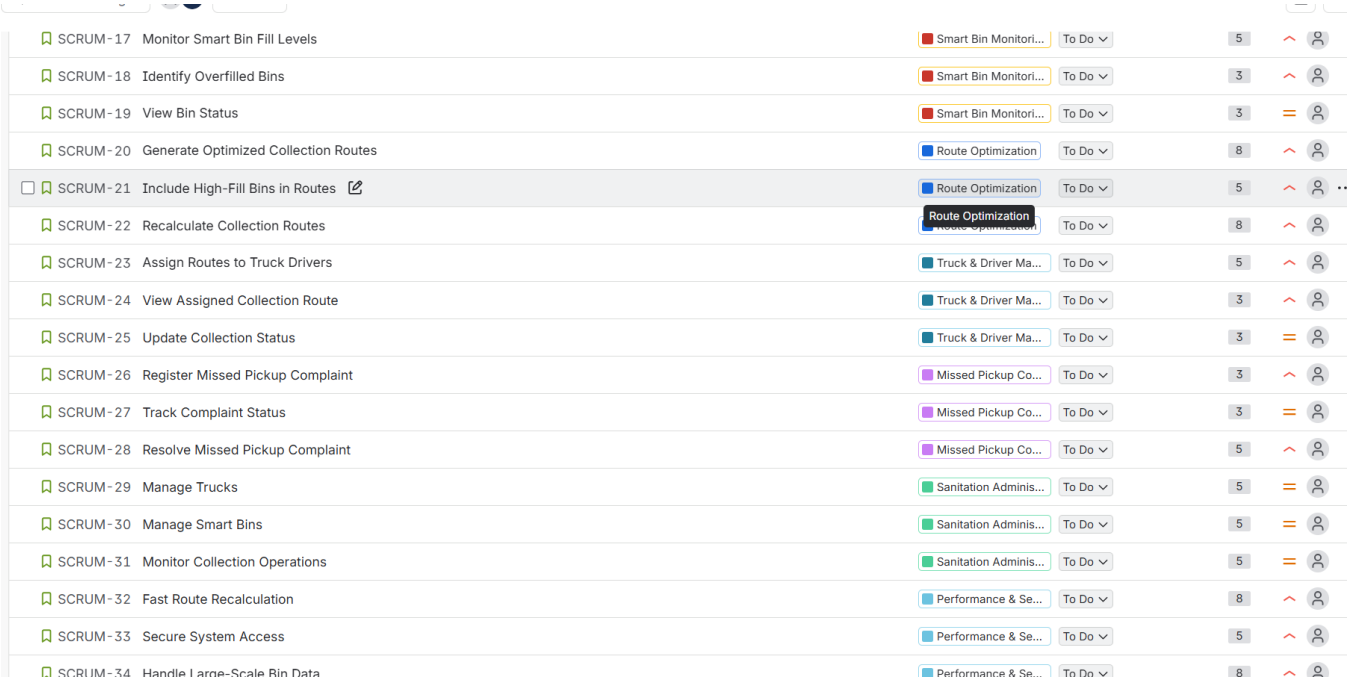
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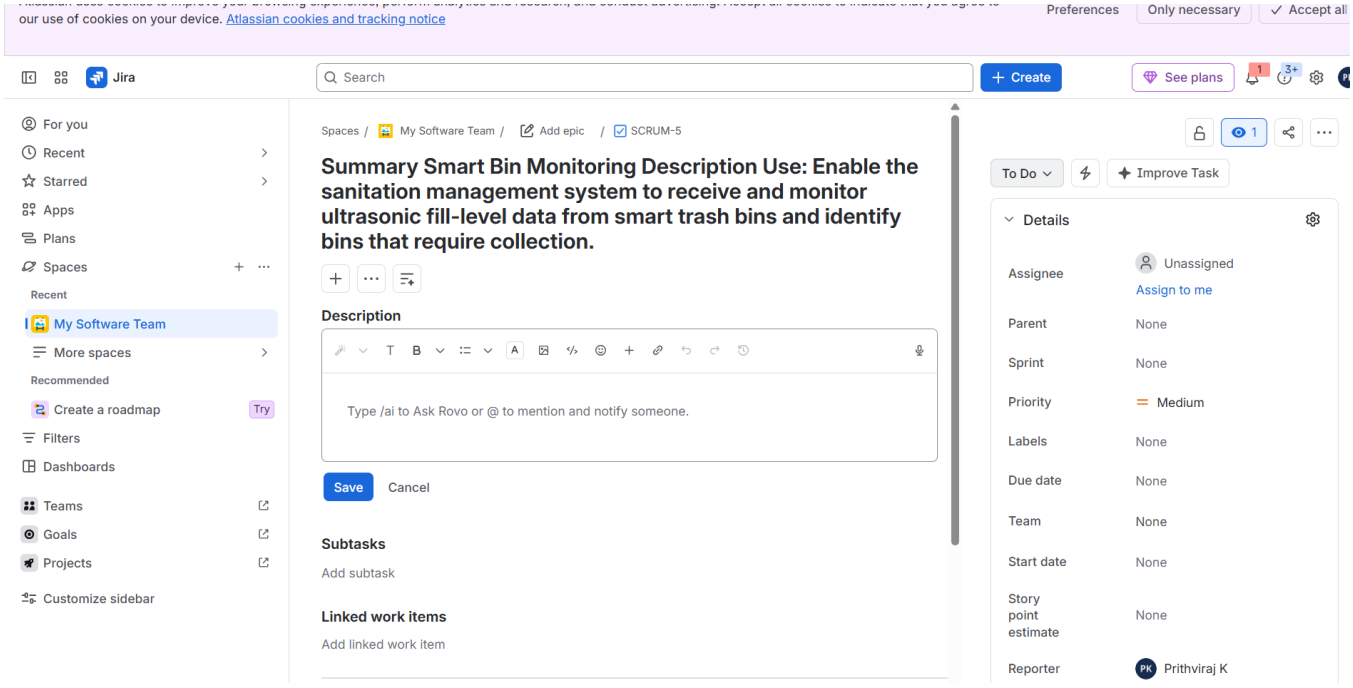
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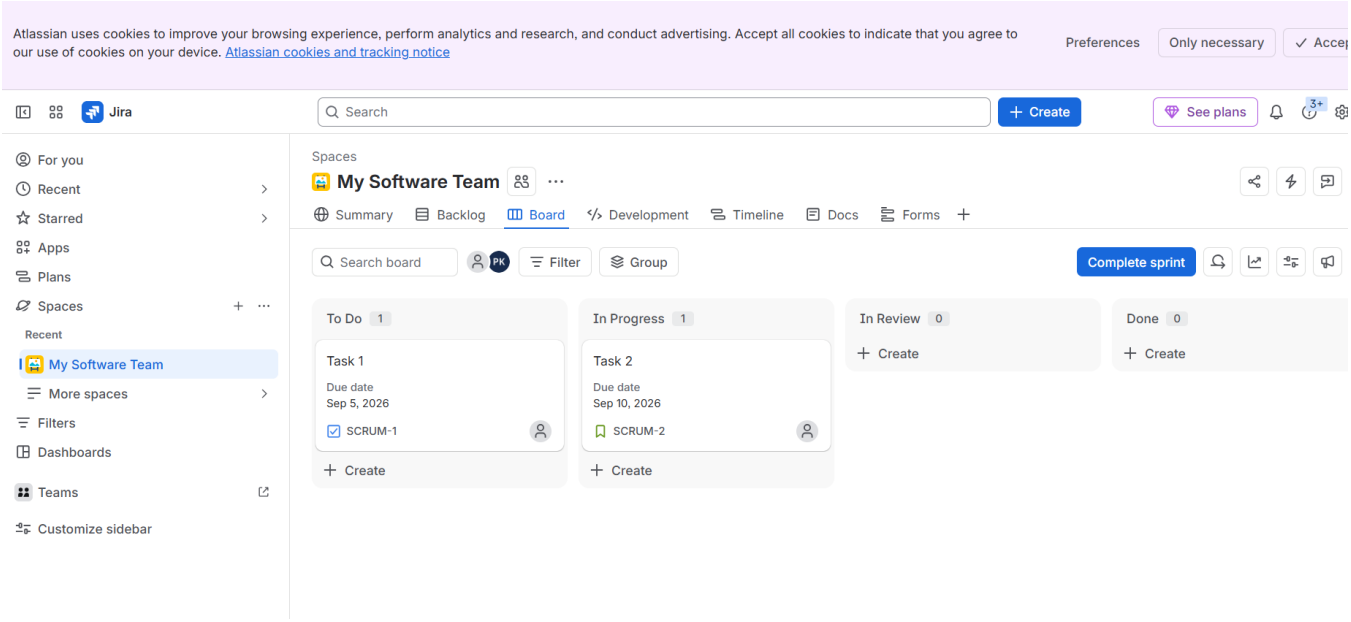
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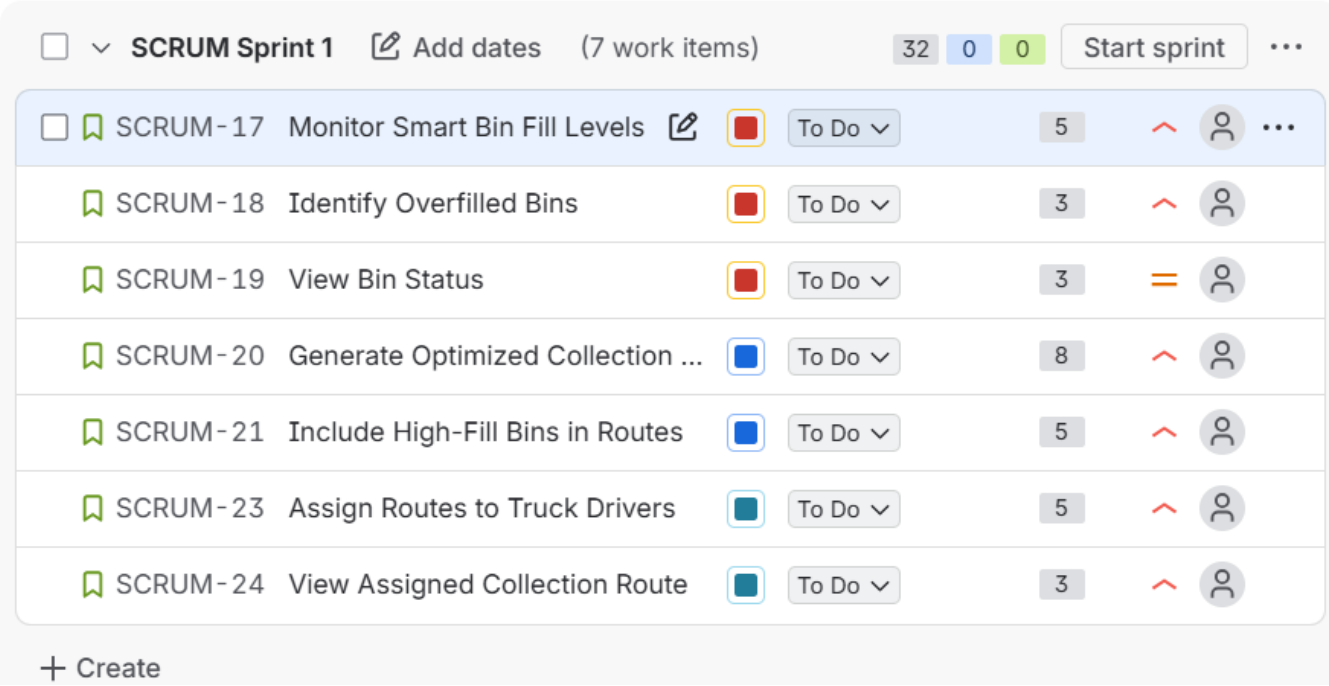
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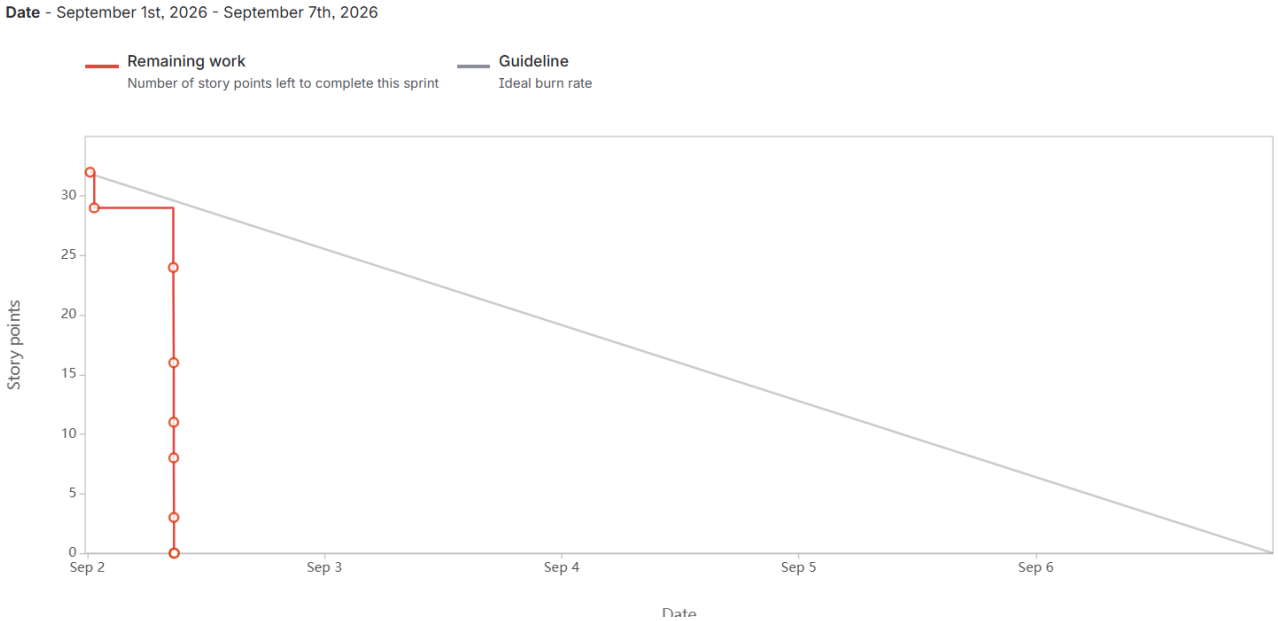
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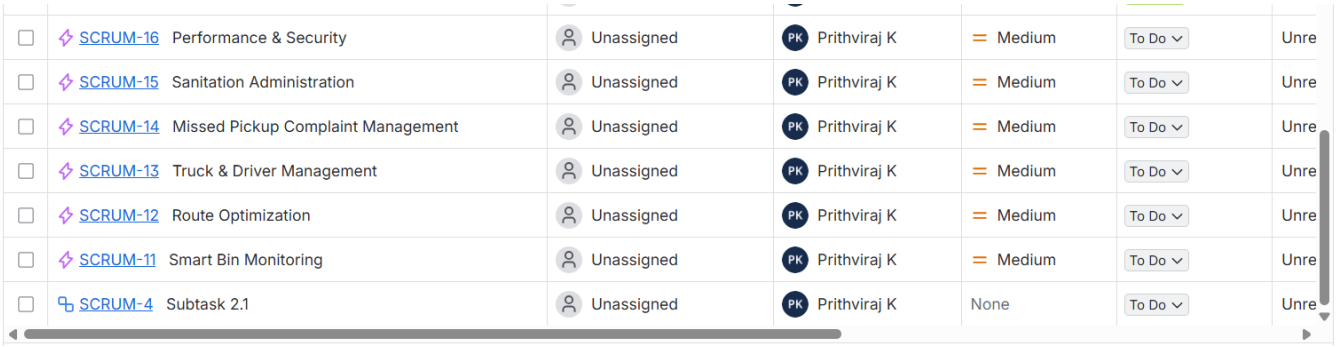
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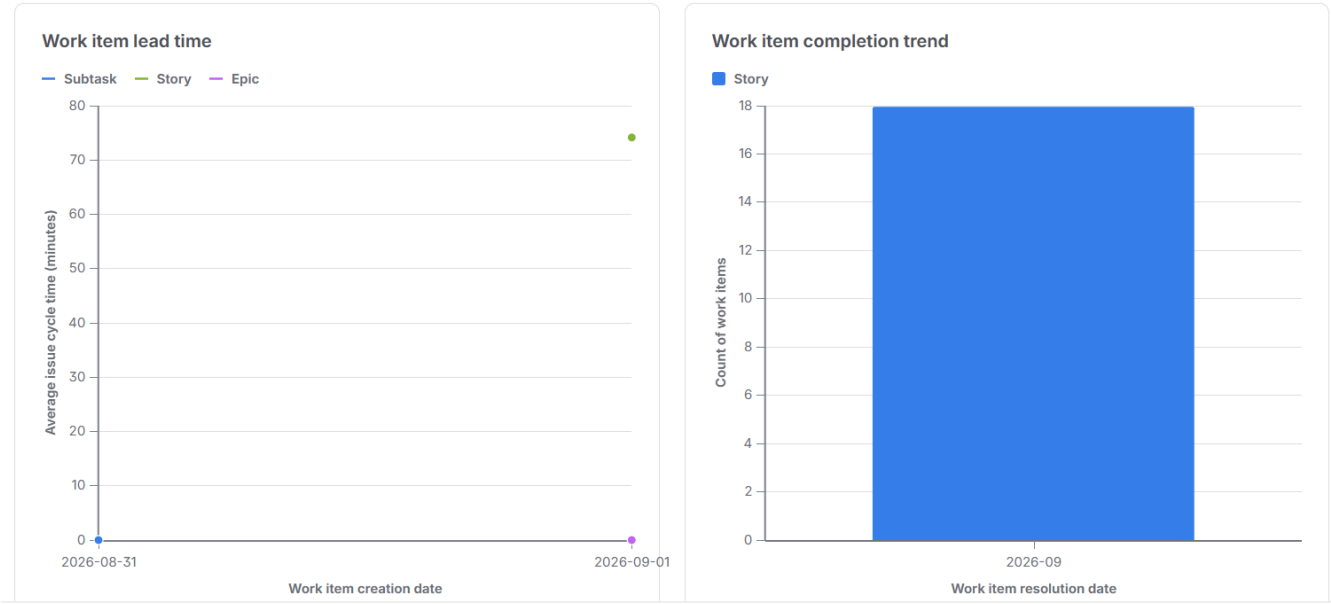


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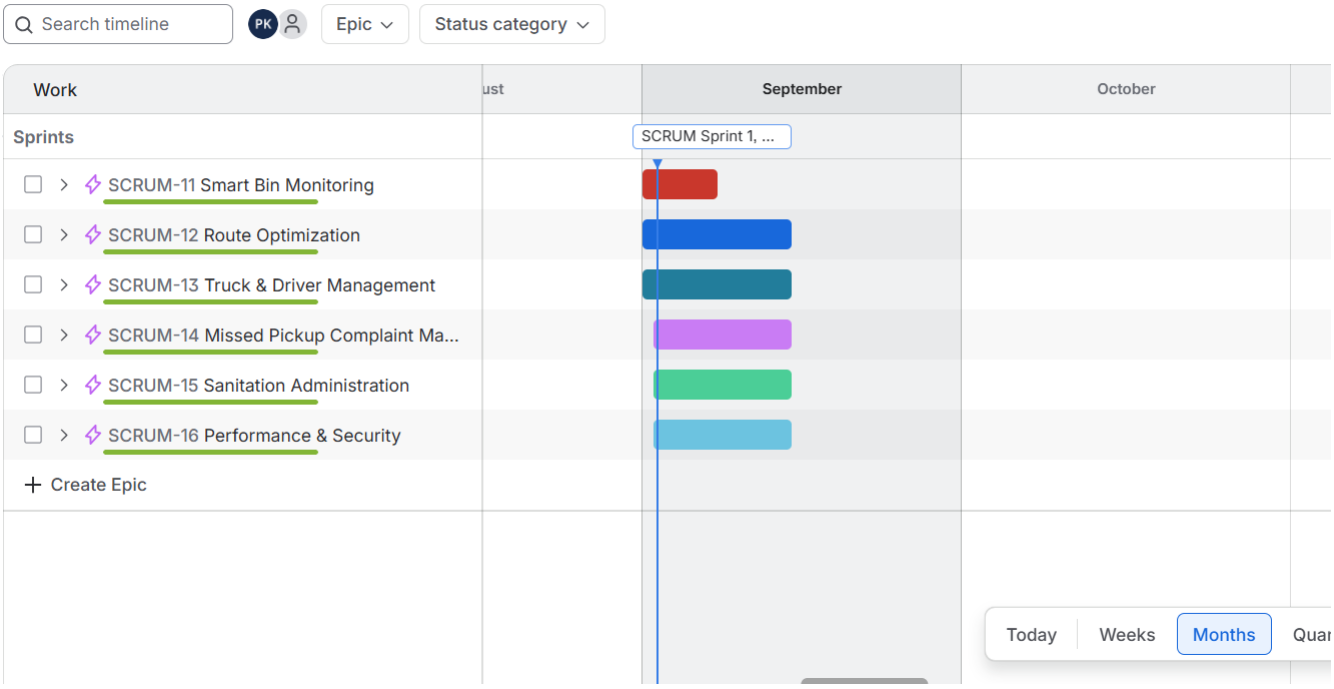


☐	SCRUM-16 Performance & Security	Unassigned	Prithviraj K	= Medium	To Do	Unre
☐	SCRUM-15 Sanitation Administration	Unassigned	Prithviraj K	= Medium	To Do	Unre
☐	SCRUM-14 Missed Pickup Complaint Management	Unassigned	Prithviraj K	= Medium	To Do	Unre
☐	SCRUM-13 Truck & Driver Management	Unassigned	Prithviraj K	= Medium	To Do	Unre
☐	SCRUM-12 Route Optimization	Unassigned	Prithviraj K	= Medium	To Do	Unre
☐	SCRUM-11 Smart Bin Monitoring	Unassigned	Prithviraj K	= Medium	To Do	Unre
☐	SCRUM-4 Subtask 2.1	Unassigned	Prithviraj K	None	To Do	Unre

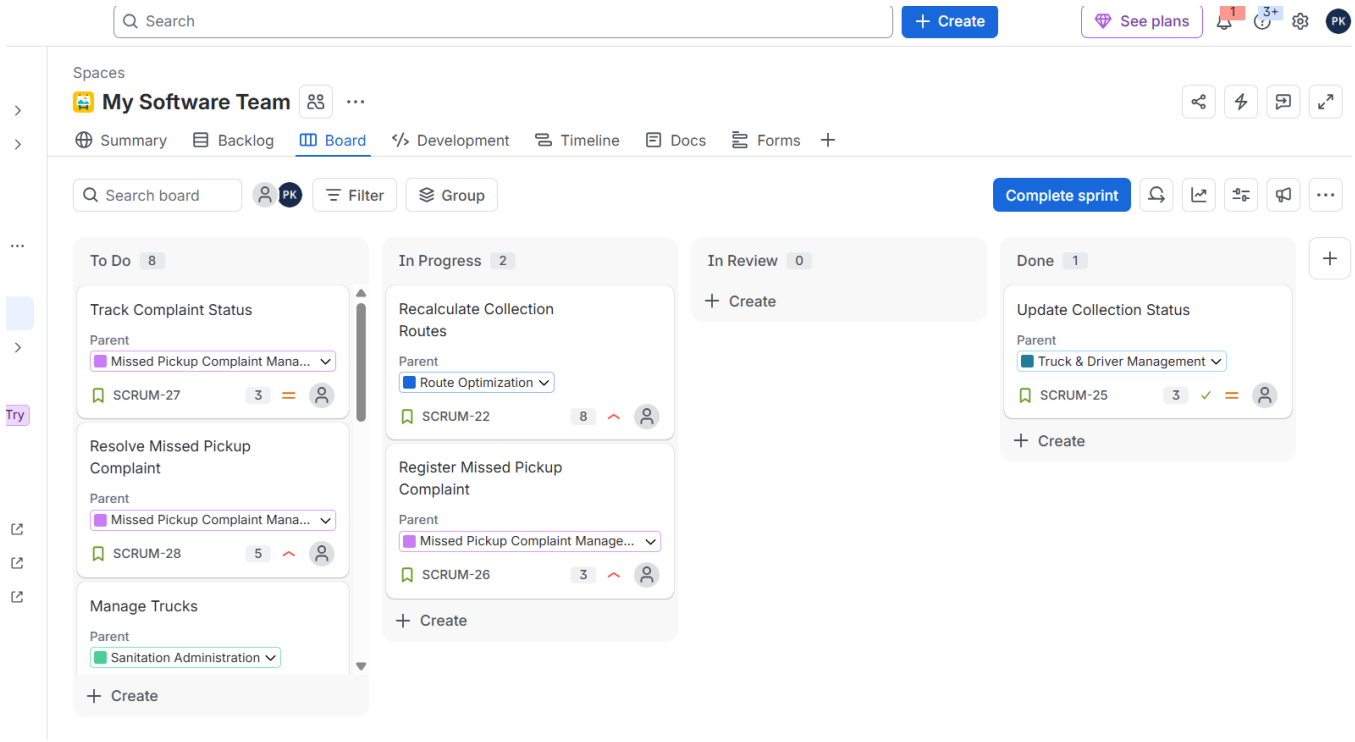
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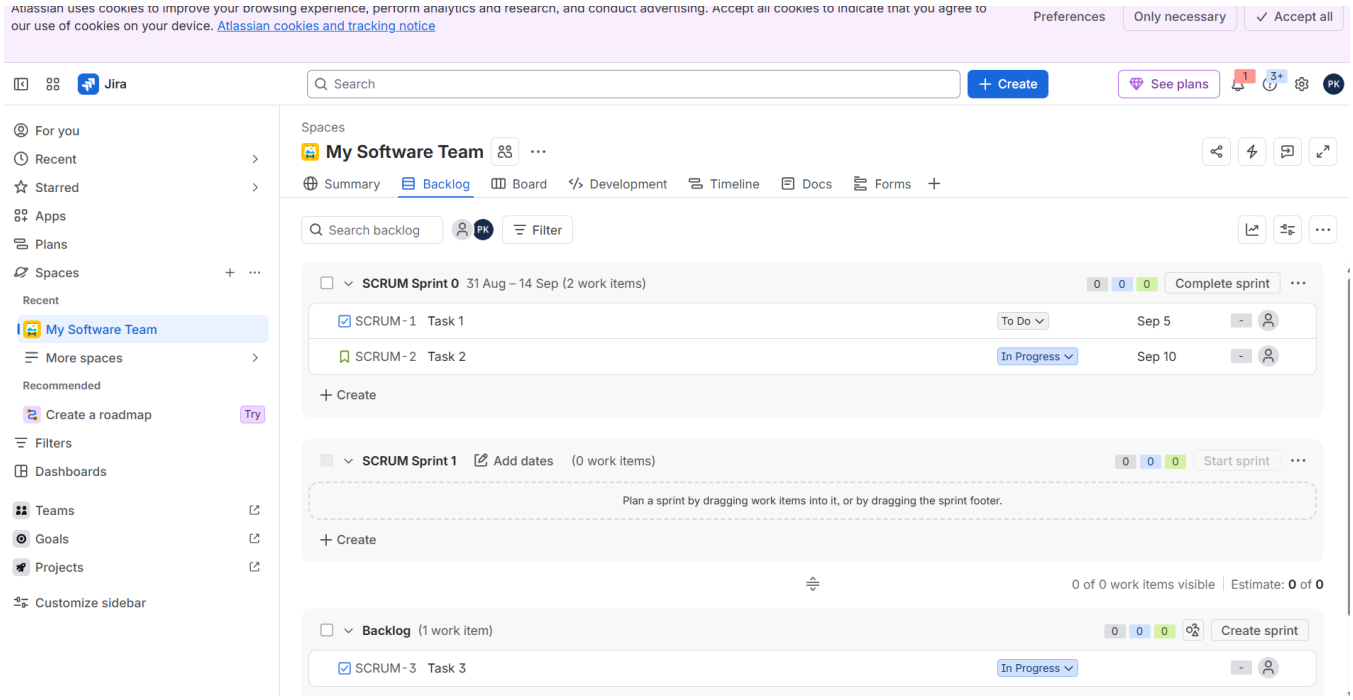
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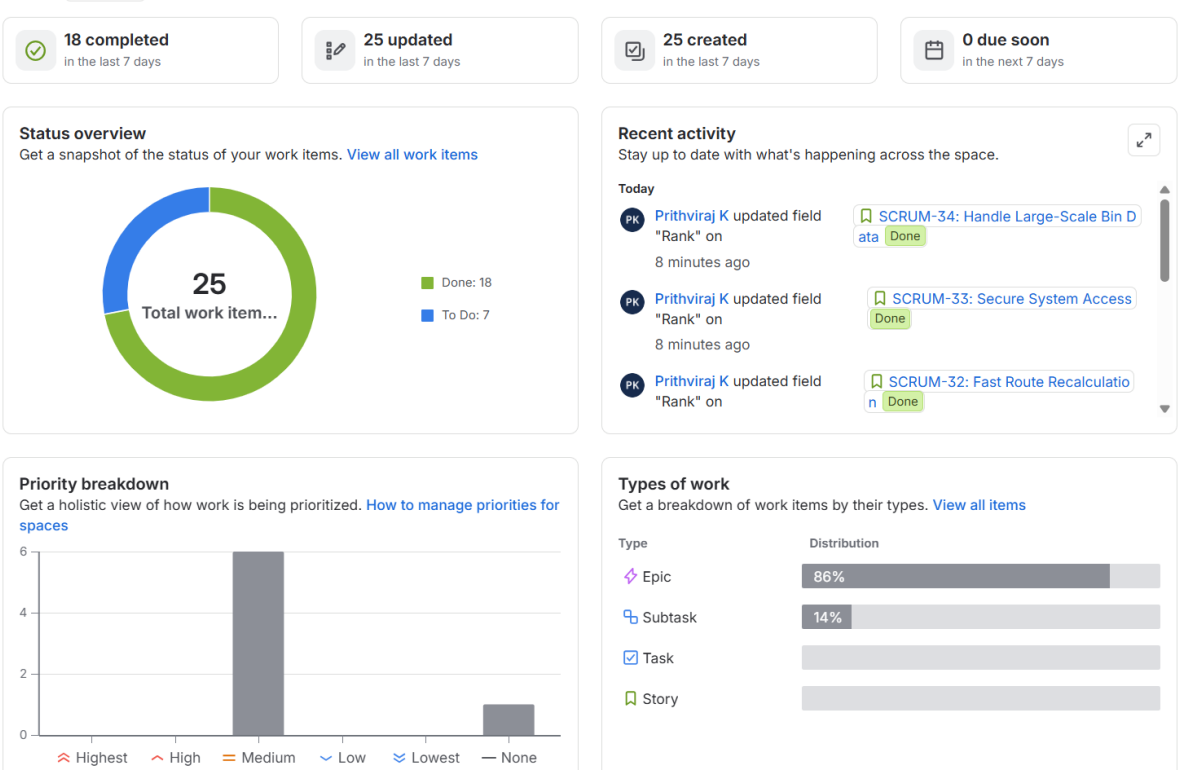
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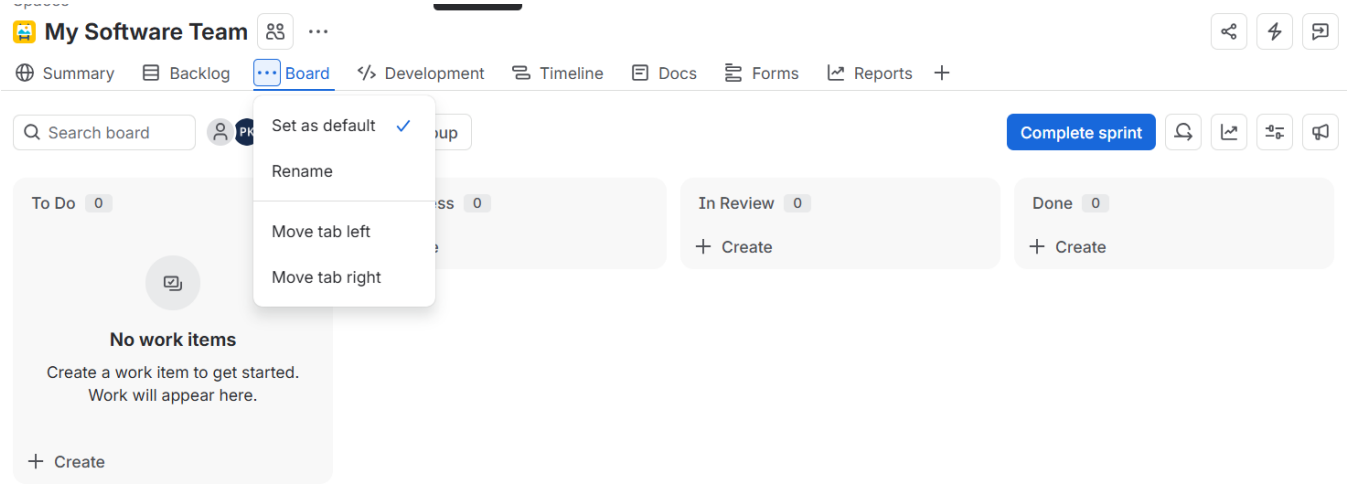
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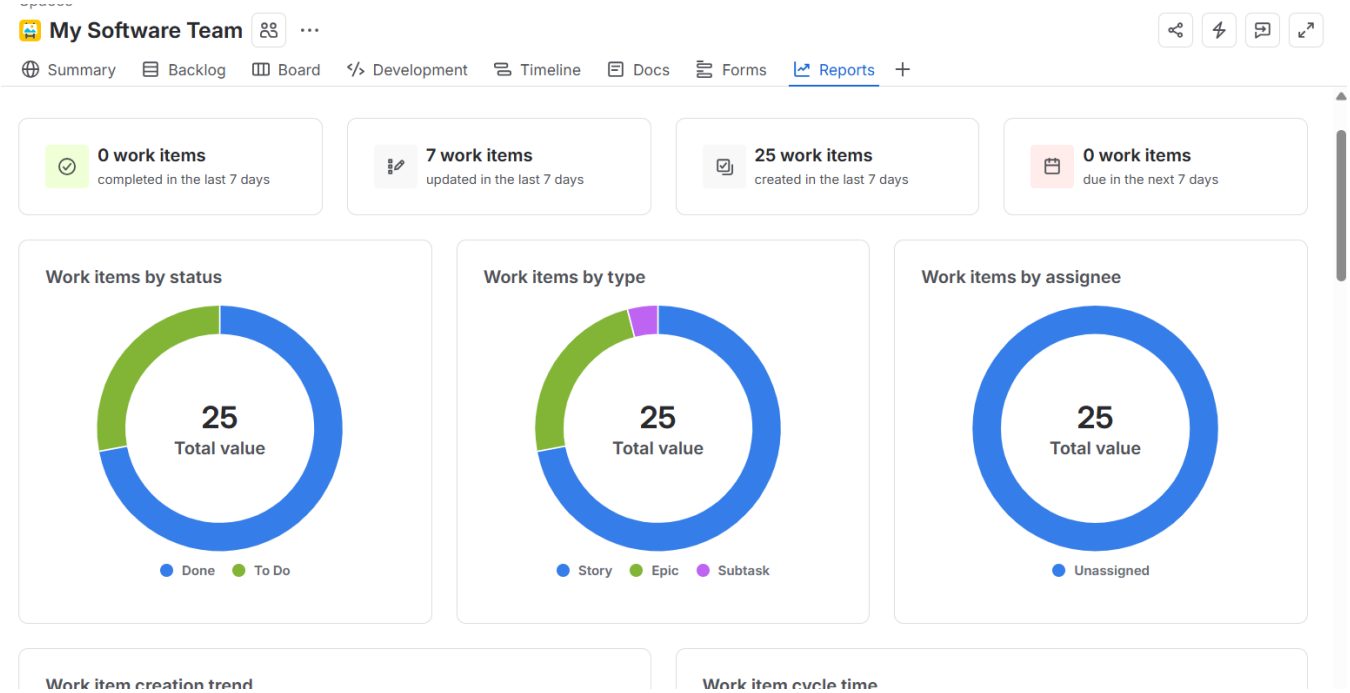
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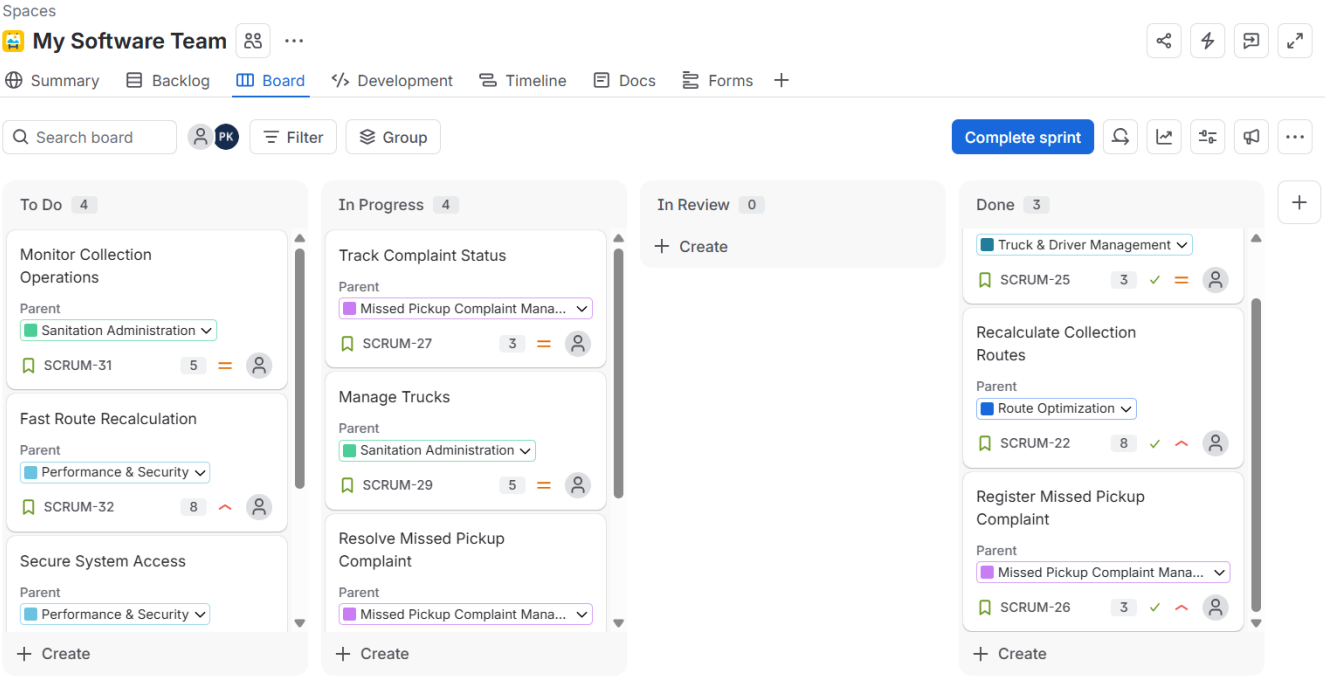
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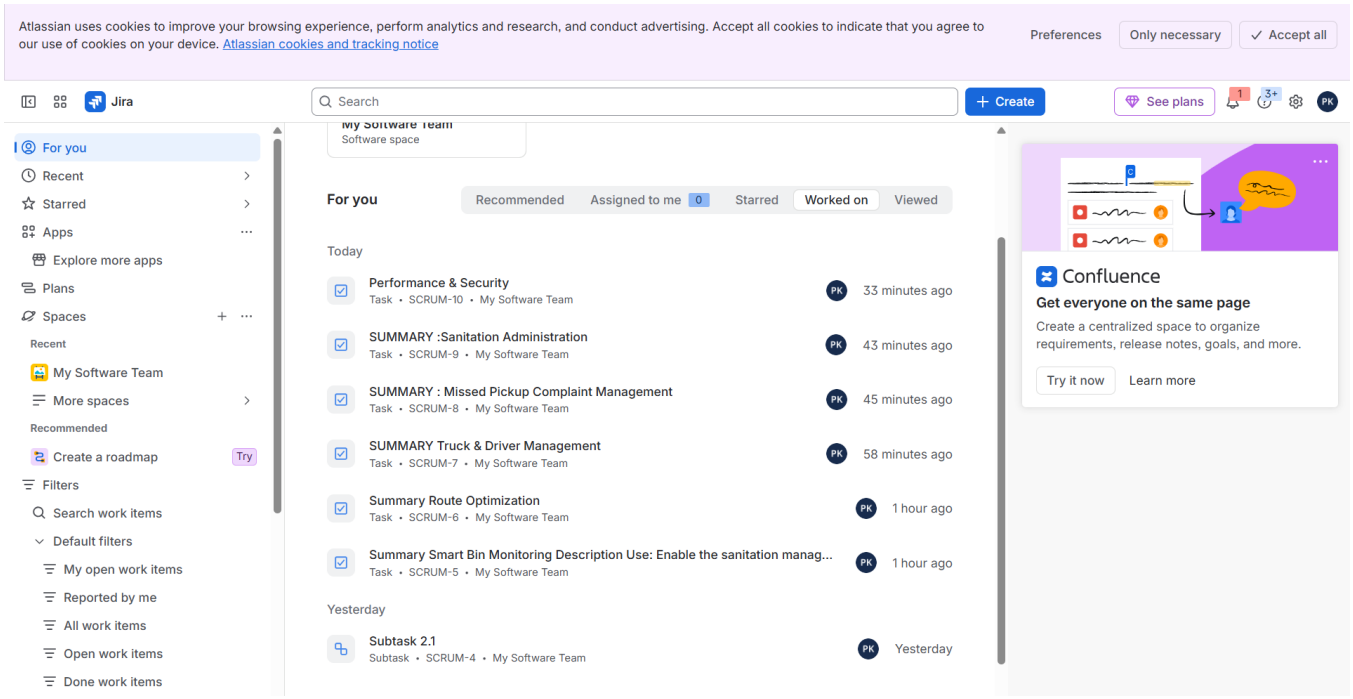
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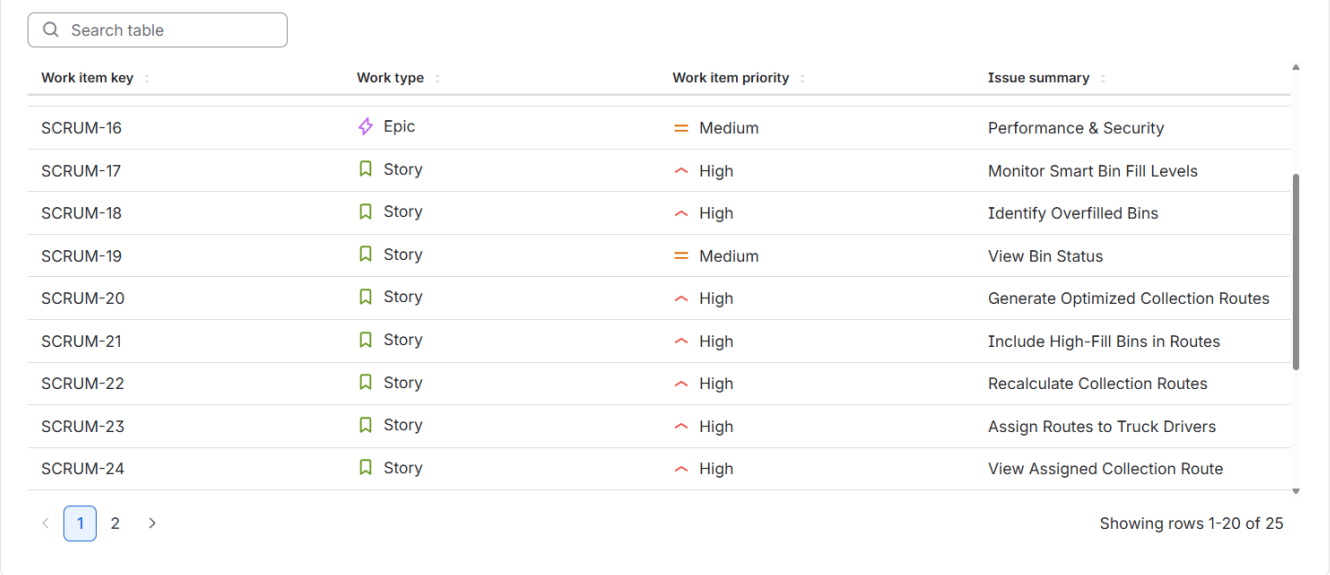
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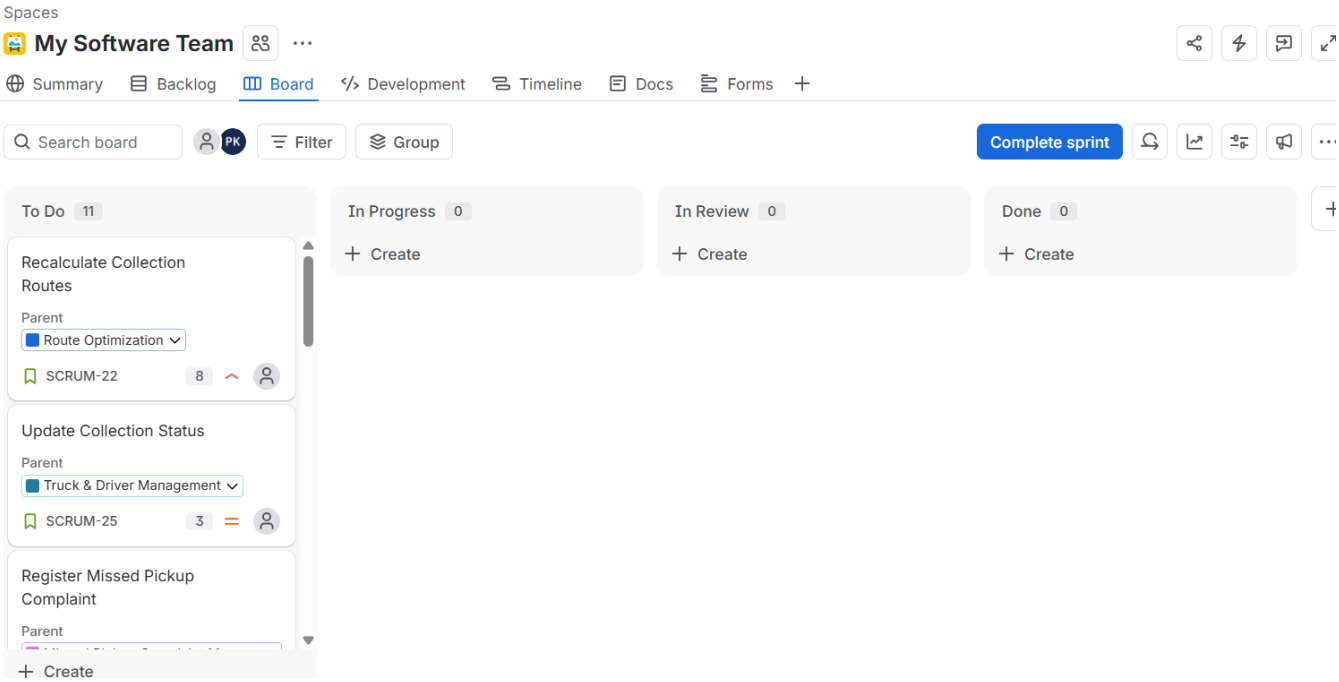
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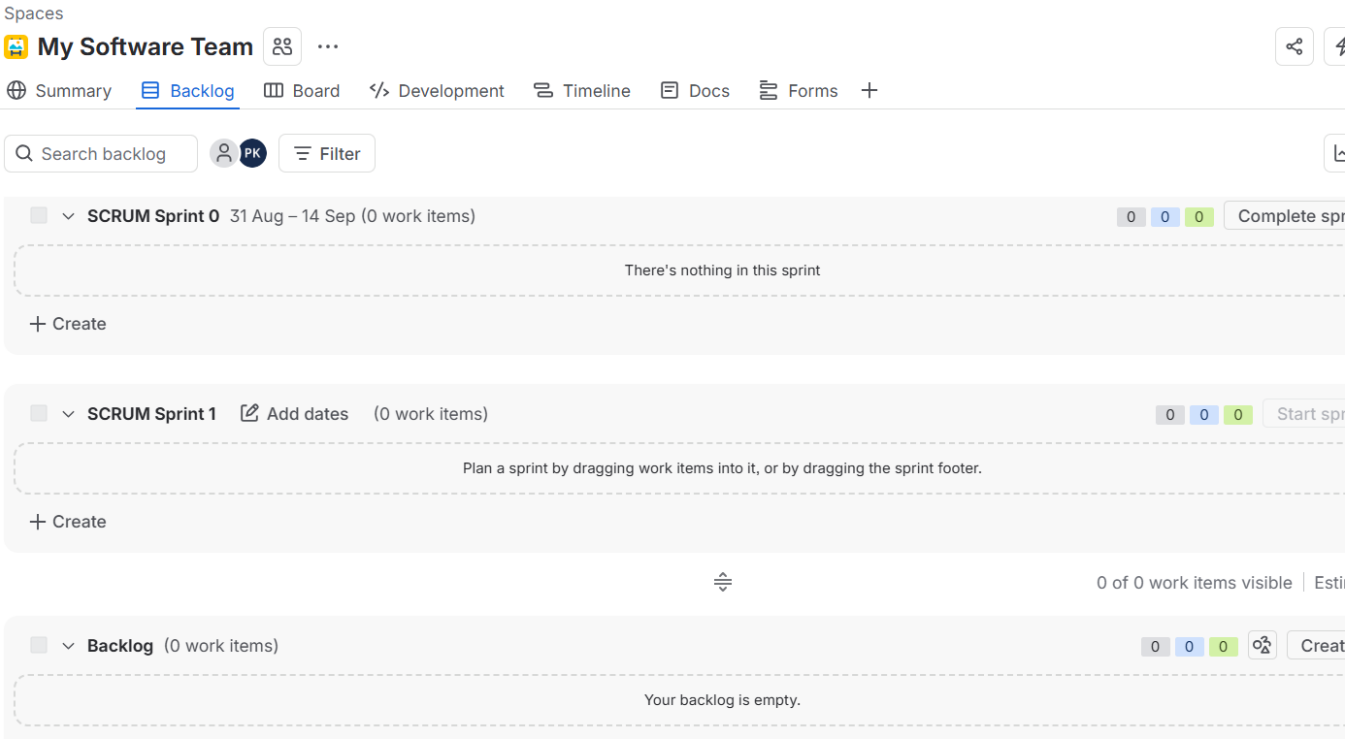
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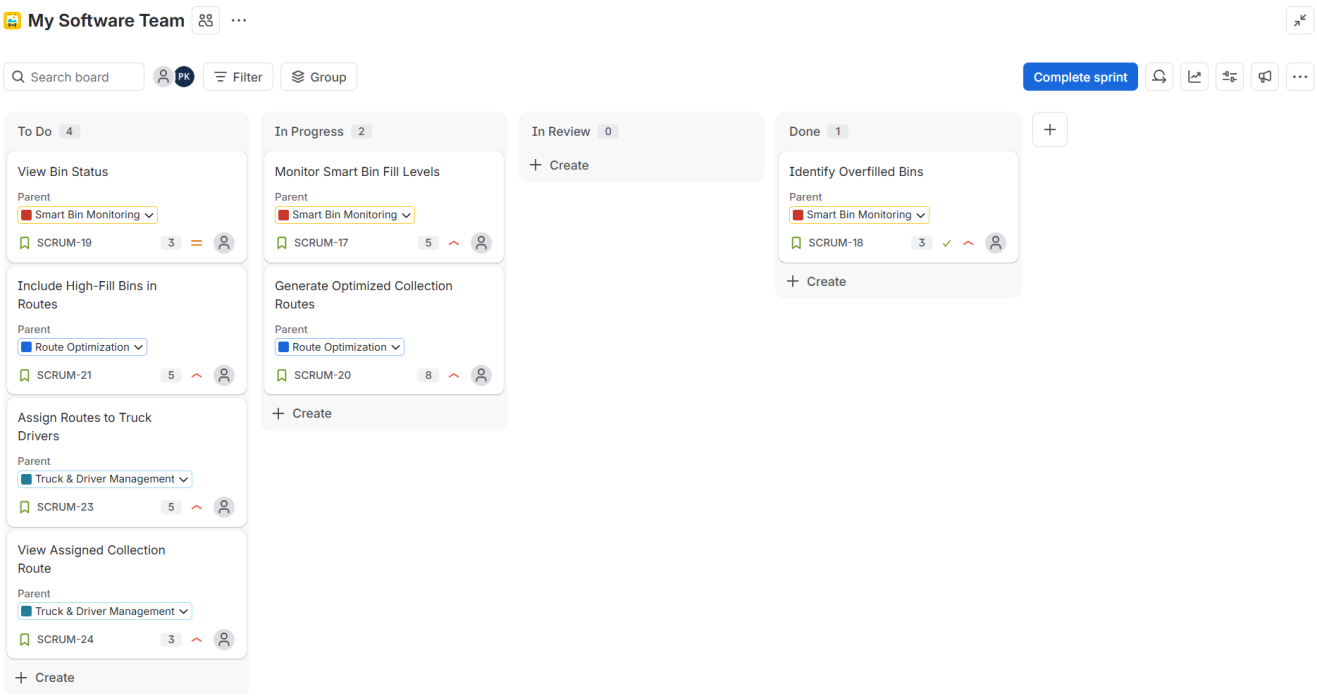
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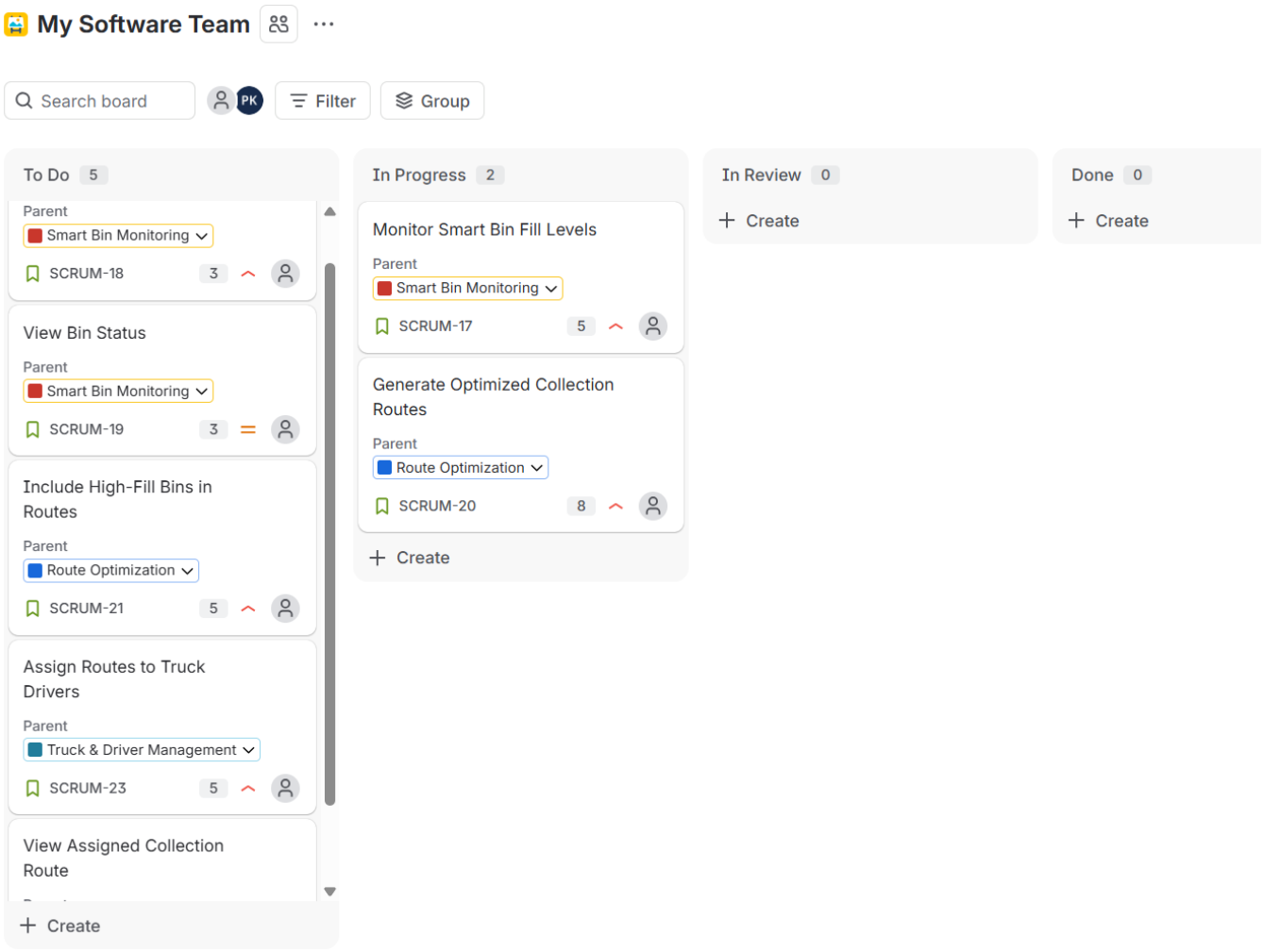
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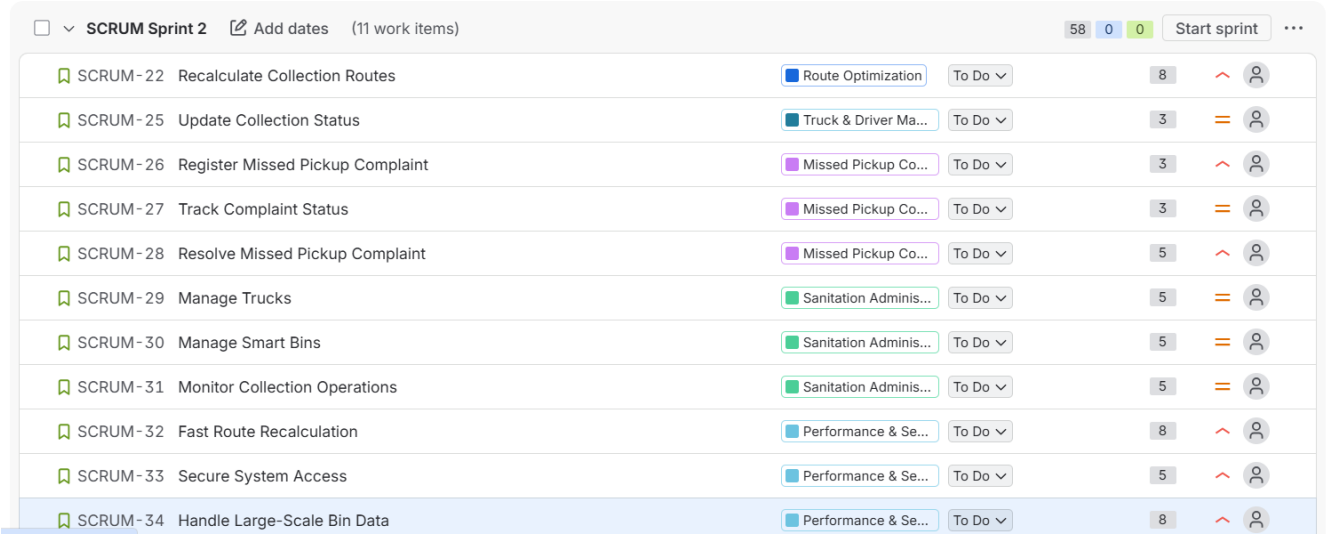
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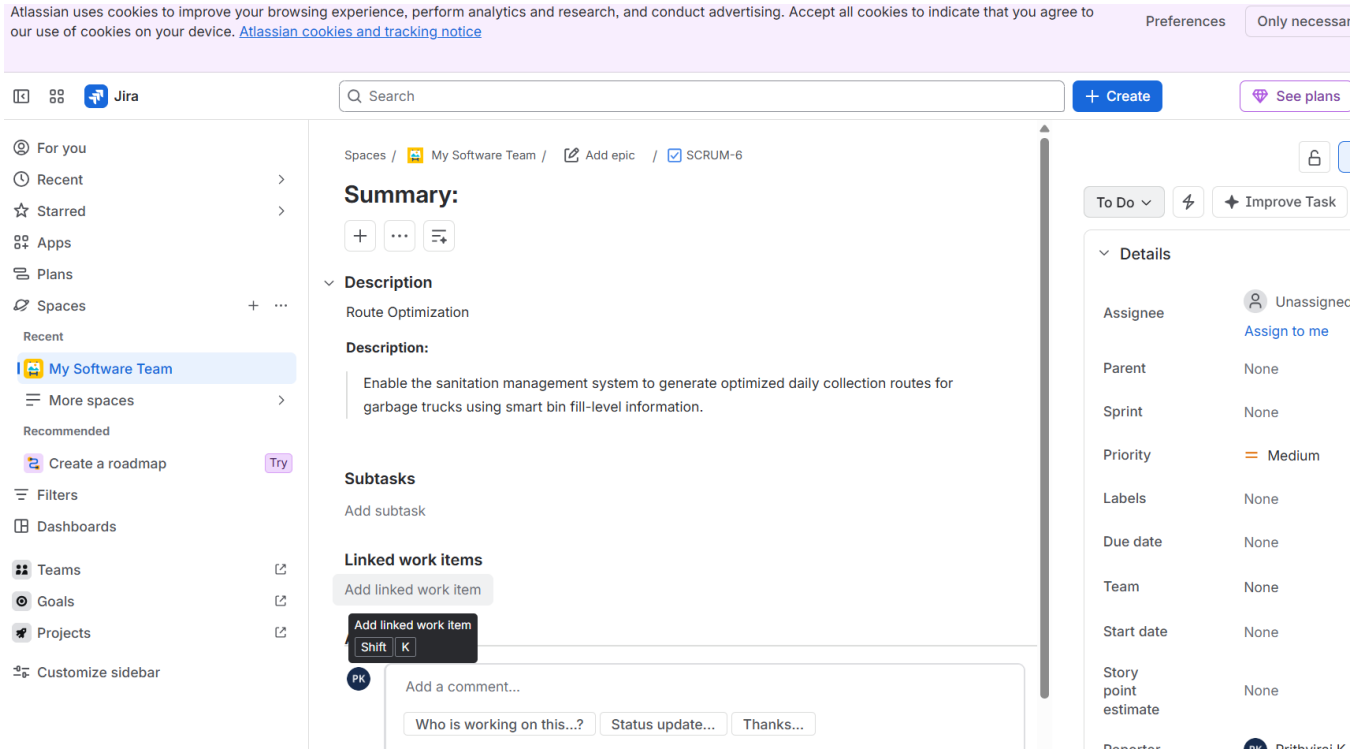
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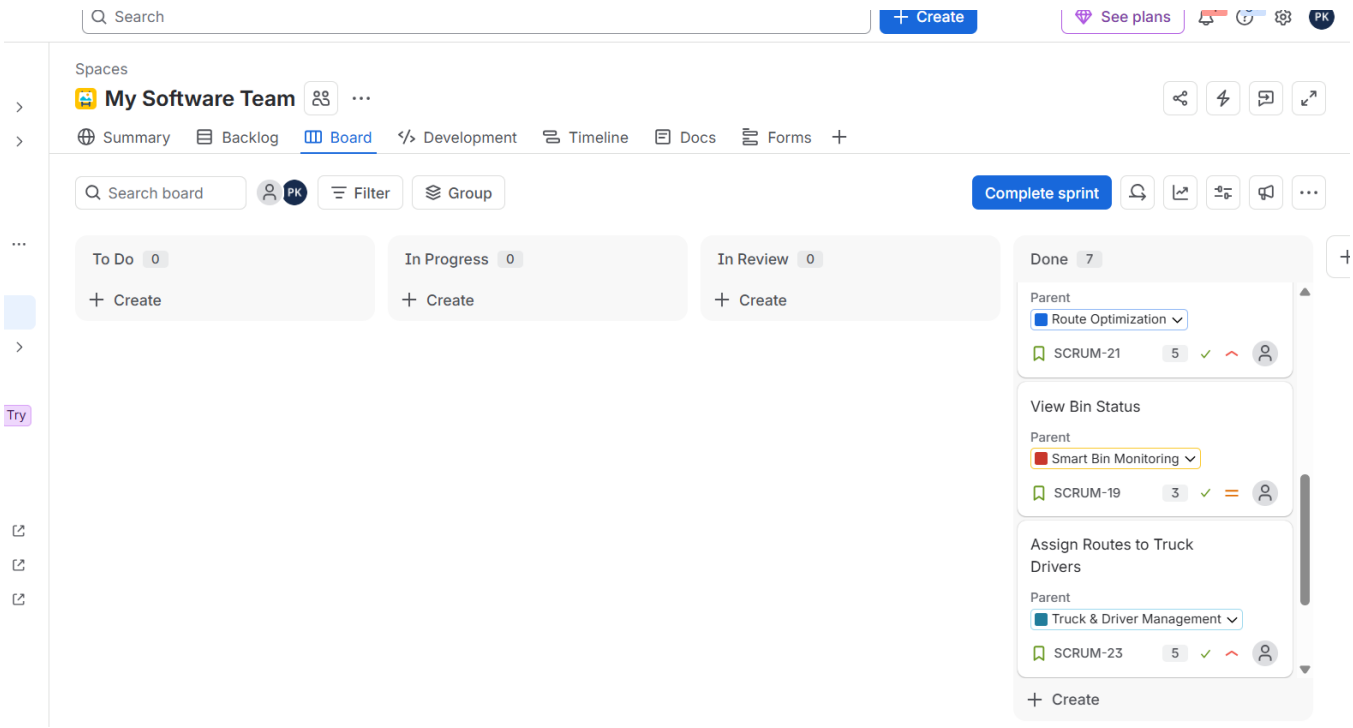
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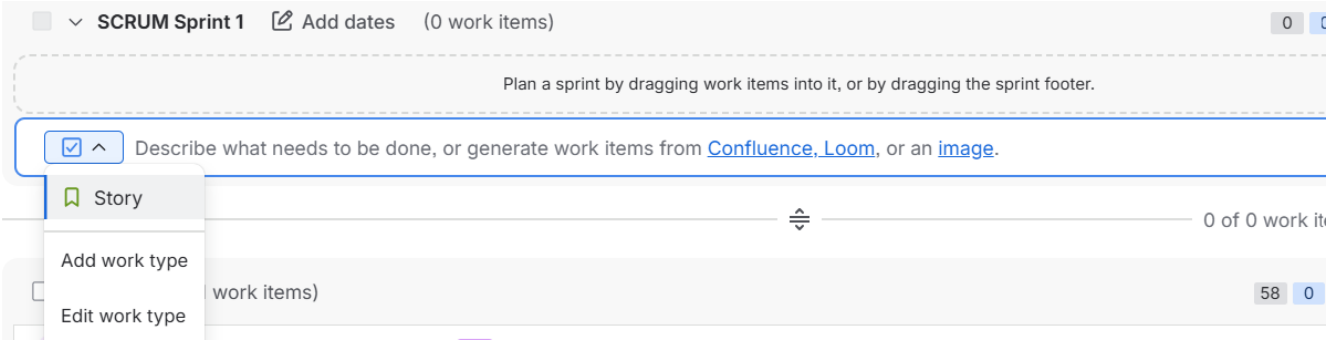
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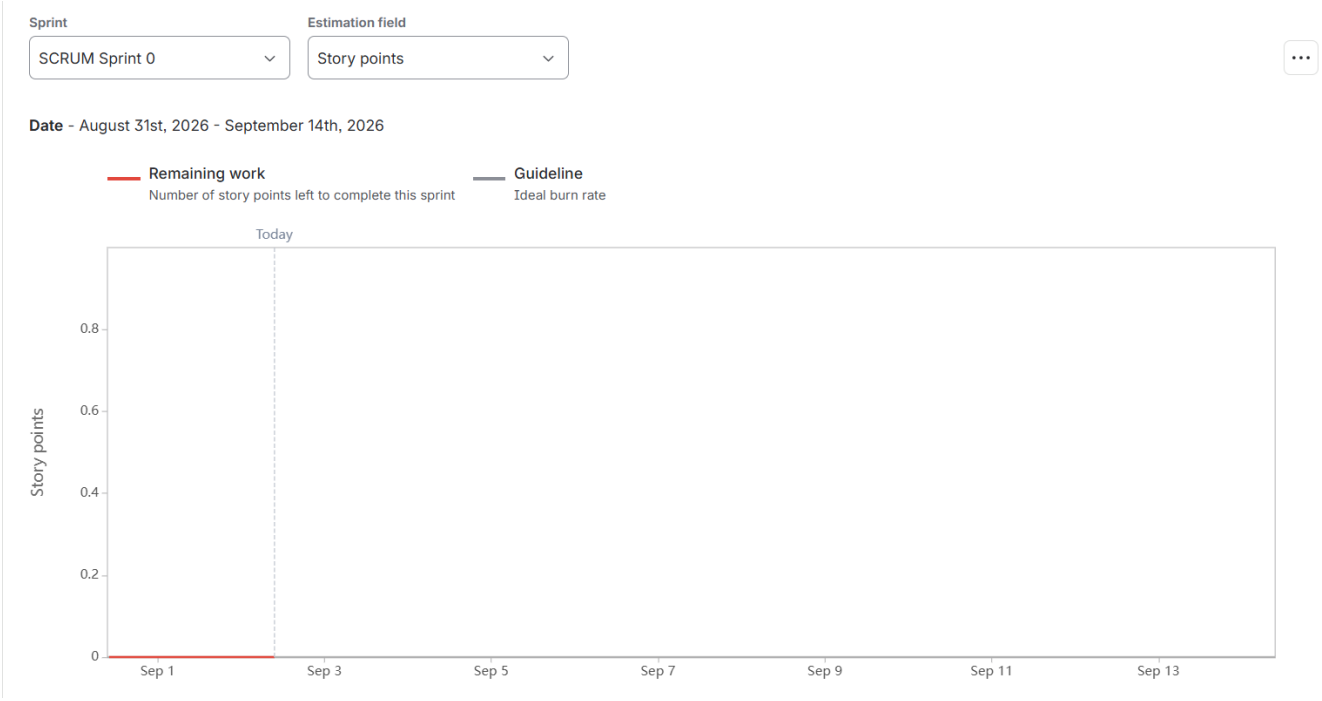
Jira screenshot 39: 193b45ad-80ca-4a4f-b4bc-a6b77a4d3bfb.png



Jira screenshot 40: 52c321b6-86d8-4fa5-96cf-5c5621daaa45.png



Jira screenshot 41: fb1071b0-99d3-41bd-a2cf-57506b78f815.png



Jira screenshot 42: 338ea9db-f9bc-4299-99a5-fcb242e46d42.png

11. Lab Deliverable Checklist

Deliverable	Status
Create Stories in Jira	Completed
Assign Story Points	Completed
Planning Poker / estimation	Completed
Prioritize & Rank Backlog	Completed
Set Priority for each story	Completed
Reorder / organize backlog	Completed
Create Sprint 1	Completed
Simulate Progress	Completed
Create Sprint 2 / finish two sprints	Completed
Add Comments	Real Jira comment screenshot still to be added
Generate Burndown Chart	Completed – Sprint 1 and Sprint 2
Summary / Reflection evidence	Completed

End of Lab 2 Submission